

RPCholesky under approximate and sketched pivot probabilities

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Abstract

Relative trace accuracy does not require vanishing sketch distortion. For $A = F^*F$, we analyze RPCholesky pivots selected from one reused oblivious sketch of F . Using an explicitly stated companion fractional-moment theorem, three weighted subspace tests and a hybrid likelihood argument give $\mathbb{E} \operatorname{tr}(A/J) \leq (1 + \varepsilon)\tau_r + \delta \operatorname{tr} A$ after $O(r/\varepsilon + r\Lambda_4(r))$ pivots at constant distortion. A suitable SparseStack moment certificate gives a row count linear in this budget plus $\log(\operatorname{tr} A/(\varepsilon\tau_r))$. We also separate the roles of approximate pivot probabilities: a lower bracket preserves scalar convergence, whereas upper domination alone admits a worst-case floor $1/a$ at every fixed finite budget. The companion two-sided bound is recalled and organized into three functional hypotheses. Further results give a spectrum-dependent frame-selection gate, good-event same-sketch reconstruction, and an independent-restart reduction whose $O(r/\varepsilon)$ consequence is conditional on a warm-start conjecture.

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1 Introduction

Let $A \in \mathbb{C}^{N \times N}$ be Hermitian positive semidefinite (PSD) with eigenvalues $\lambda_1 \geq \dots \geq \lambda_N \geq 0$, and put $\tau_r(A) = \sum_{j>r} \lambda_j$, the trace error of the best rank- r approximation. Randomly pivoted Cholesky starts at $R_0 = A$. At a nonzero residual R it chooses a coordinate i with probability

$$\pi_i(R) = \frac{R_{ii}}{\text{tr } R} \quad (1)$$

and replaces R by the Schur complement

$$R^{(i)} = R - \frac{R e_i e_i^* R}{R_{ii}}. \quad (2)$$

After k pivots with index set J the approximation is $A - R_k$ and its trace-norm error is $\text{tr } R_k$. Chen, Epperly, Tropp and Webber [1] give the modern account of the method and its connection with adaptive sampling [3, 4]. The paper “Iterated-logarithmic pivot bounds for randomly pivoted Cholesky” [8] proves $\mathbb{E} \text{tr } R_k \leq (1 + \varepsilon) \tau_r(A)$ after $O(r/\varepsilon + r \log \log \log \log r)$ pivots, uniformly over all inputs; Epperly [5] proves a count $r/\varepsilon + 2r\sqrt{\log r} + r \log(1/\varepsilon) + 2.3r$.

In practice the pivot law is often not exactly (1). The residual diagonal may be maintained approximately, or the method may run on a compressed surrogate of A . This paper asks which parts of the pivot-count theory survive a perturbation of the pivot law, and proves a precise answer in two settings: abstract multiplicative perturbations and sketching.

1.1 Main results

Throughout, a *pivot rule* chooses, conditionally on the complete past (which may include external randomness), a coordinate with law q supported on the positive diagonal entries of the current residual, and then performs the exact Schur update (2). We write π for the exact law (1) at the same state and $0 < a \leq 1$ for a comparison factor.

Theorem 1.1 (Robustness trichotomy). *Let $r \geq 1$, $0 < \varepsilon \leq 1$ and $\tau = \tau_r(A) > 0$.*

- (i) *Two-sided. If $a\pi_i \leq q_i \leq \pi_i/a$ at every step, then $\mathbb{E} \text{tr } R_k \leq (1 + \varepsilon)\tau$ for every $k \geq \min\{N, K_a\}$, where for $r \geq 2$, $K_a = r + g_a + s_a + n_a$ is the explicit count of Corollary 3.3; rank one uses Remark 3.4. With the parameter $\beta \in (0, 1)$ of the imported moment (Theorem 2.5) held fixed,*

$$K_a = O\left(\frac{r}{a\varepsilon} + \frac{r}{a} \log \frac{e}{a} + r\Lambda_4(r)\right),$$

where Λ_4 is the fourfold iterated logarithm of (3).

- (ii) *Lower only. If $q_i \geq a\pi_i$ at every step, then from any state with $\mathbb{E} \text{tr } R_{k_0} \leq (1 + u)\tau$ and $u > \varepsilon$, a further $\lceil (r/a)(F(\varepsilon) - F(u)) \rceil$ pivots give $\mathbb{E} \text{tr } R \leq (1 + \varepsilon)\tau$, where $F(u) = u^{-1} - \log u$. In particular $\lceil (r/a)(\varepsilon^{-1} + \log \varepsilon^{-1} + \log(\text{tr } A/\tau)) \rceil$ pivots suffice from the start. There is no accuracy floor.*
- (iii) *Upper only. If $q_i \leq \pi_i/a$ at every step and $r \geq 2$, then $\mathbb{E} \text{tr } R_k \leq (10/a + 1/r)\tau$ for $k \geq r + g_a + s_a$. No multiplicative accuracy below $1/a$ is possible: for every r , every budget k and every $\delta > 0$ there are a diagonal matrix and a memoryless rule with $q \leq \pi/a$ for which $\mathbb{E} \text{tr } R_k \geq (1/a - \delta)\tau_r(A)$.*

Part (i) is Corollary 3.3, part (ii) is Theorem 4.1, and part (iii) is Theorem 4.2. All three follow from Proposition 3.1, a conversion that isolates three functional inequalities: an initial moment, an upper comparison of one determinant functional, and a lower comparison of the expected trace decrease. That formulation is what makes the sketched theorem possible, because a sketch provides these three inequalities without providing a per-coordinate bracket.

For the second setting let $F \in \mathbb{C}^{D \times N}$ and $A = F^*F$. Sketched RPCholesky draws a random $\Pi \in \mathbb{C}^{m \times D}$, independent of the pivoting, and runs the pivot process on $B = (\Pi F)^*(\Pi F)$. It returns the index set J , and the error is measured on A . We say that Π has the *moment property* $\text{MP}(d, \ell, \varepsilon_0)$ if every isometry U with at most d columns satisfies $\mathbb{E} \text{tr}(U^* \Pi^* \Pi U - I)^{2\ell} \leq d \varepsilon_0^{2\ell}$.

Theorem 1.2 (Sketched RPCholesky). *Let $r \geq 2$, $0 < \varepsilon \leq 1$, $0 < \eta \leq 1/4$ and $0 < \beta < 1$, and put*

$$\gamma_\eta = \frac{1 - \eta - \eta^2}{1 + \eta}, \quad \chi_\eta^2 = \frac{(\eta/\sqrt{2} + \eta^2/(2(1 - \eta)))^2}{(1 - \eta - \eta^2)^2}.$$

Let $K \geq K^{\text{sk}}$, the explicit count (19), which is $O(r/\varepsilon + r\Lambda_4(r))$ for fixed η and β . If Π has $\text{MP}(K + 1, \ell, \eta/2)$ with $\ell \geq 2$, then K steps of sketched RPCholesky satisfy

$$\mathbb{E} \text{tr} R_A(J) \leq (1 + \varepsilon)\tau_r(A) + P_{\text{fail}} \text{tr} A, \quad P_{\text{fail}} \leq 2(K + 1)e^{K\chi_\eta^2/2}2^{-\ell/2}.$$

With probability at least $1 - P_{\text{fail}}$ the same sketch also gives a reconstruction $F_J(\Pi F_J)^+ \Pi F$ whose squared Frobenius error is at most $(1 + \eta^2/(2(1 - \eta)^2)) \text{tr} R_A(J)$.

This is Theorem 5.8 and Corollary 5.12. The accuracy parameter ε is paid in pivots, through the factor $1/\gamma_\eta$ of the robust count, and never in the sketch distortion. Taking ℓ of order $K\eta^2 + \log(K \text{tr} A/(\varepsilon\tau_r))$ makes $P_{\text{fail}} \text{tr} A \leq \varepsilon\tau_r$. For SparseStack this means

$$m = O\left(\frac{K + \log(\text{tr} A/(\varepsilon\tau_r))}{\eta^2}\right)$$

rows, with no dependence on N , on D , or on a union bound over coordinates (Corollary 5.15). For two-round SRHT the quadratic moment term gives $M = O(K/\eta^2 + K^2\eta^2 + \eta^{-2} \log^2(K \text{tr} A/(\varepsilon\tau_r)))$. A transfer that compares the sketched and exact path laws only through their χ^2 divergence would certify the same accuracy through the usual variance estimate only at the scale $m \gtrsim K/\varepsilon^2$; the hybrid argument avoids that cost because it never compares terminal errors directly (Section 5.8).

Two further results use the same likelihood calculus. Theorem 6.1 prices the selection of a fixed-dictionary frame by full-rank RPCholesky with the pointwise factor $\Lambda_C = k! \det C / \prod_{j < k} \tau_j(C) \in [1, k!]$, and Proposition 6.2 gives a Rényi price that wins only for $k \gtrsim 128$. Theorem 7.1 shows that $\lceil 1/\theta \rceil$ independent chains and their union satisfy $\mathbb{E} \text{tr}(A/S) \leq (\mathbb{E} T^\theta)^{1/\theta}$, so a fractional warm-start moment would give an $O(r/\varepsilon)$ restart variant. On the scale-separated family whose first moment grows like 2^r , exact enumeration gives a square-root moment within 0.3% of one.

1.2 Organization

Section 2 collects the deterministic lemmas and the imported moment theorem. Section 3 proves the functional conversion phase by phase and the two-sided count. Section 4 proves the lower-only and upper-only statements, including the floor construction. Section 5 treats

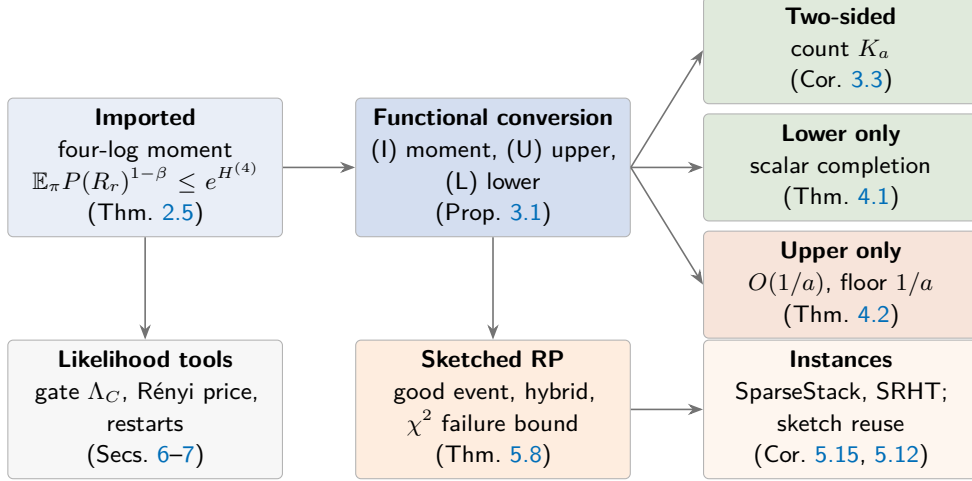


Figure 1: Roadmap. The imported initial moment and the functional conversion feed every pivot-count statement. The sketched theorem enters the conversion through the same three inequalities as the two-sided bracket, with $a = \gamma_\eta$.

sketched RPCholesky. Section 6 gives the fixed-dictionary gate and Section 7 the restart reduction. Section 8 contains the numerical illustrations and Section 9 the open problems.

1.3 Prior work and the contribution of this paper

Chen, Epperly, Tropp and Webber [1] established the modern RPCholesky analysis, including the exact conditional trace decrease. The adaptive-sampling and volume-sampling arguments originate with Deshpande and collaborators [3, 4]. Our scalar phase and pivot-product identities build on those arguments. The initial fractional determinant moment is imported from [8]; this paper does not claim a new proof of the exact-pivoting four-log bound. The two-sided count itself is already Proposition 8.1 of that companion paper. We reproduce it to expose the functional hypotheses needed below.

Divan and Gilles [2, Theorem 3.1] already analyze upper-dominated laws $q \leq \pi/\gamma$ through elementary symmetric polynomials. Their sketchy RPLU analysis uses a fresh block at each iteration. The contribution here is the functional reformulation of the companion conversion, the complete one-sided trichotomy, and the finite-budget floor under upper domination alone. For sketching, we analyze RPCholesky on the Gram matrix of one reused sketch and measure its selected-column error on the original matrix. Weighted tests replace a coordinate union bound, and the hybrid chain transfers the drift inequalities while charging failures through a likelihood second moment.

Epperly’s analysis of adaptive randomized pivoting [6] already identifies full-rank pivoting on an orthonormal frame with volume sampling. Accordingly, the case $\Lambda_C = 1$ of our frame gate is a known identity, and its $k!$ bound is the comparison of [4]. Our refinement retains the residual spectral tails in Λ_C and applies the resulting price to sketch-assisted selection from a fixed dictionary. Union amplification is an elementary independent-restart reduction; its $O(r/\varepsilon)$ consequence remains conditional on a uniform warm-start moment. These distinctions are the basis of our novelty claims relative to the works cited here.

Scope of the guarantees. The main trace bound concerns A/J , equivalently the best projection onto the selected original columns. Reconstructing with the same sketch has a separate good-event guarantee; it need not have controlled error on failed histories. The explicit embedding constants are imported from repository manuscripts [9, 10]. Their use here is conditional on those stated certificates, with each required estimate restated in Section 5.

2 Preliminaries and the imported moment

All logarithms are natural. For a PSD matrix R we write $\tau_j(R)$ for the sum of its eigenvalues beyond the j th. For an index set J we write $A/J = A - A_{:J}A_{JJ}^+A_{J:}$ for the Schur complement (residual) of A after pivoting on J . Positive pivots never repeat an index, and A_{JJ} is then invertible, so A/J does not depend on the order of the pivots. Positive versions of the iterated logarithms are

$$\Lambda_1(r) = \log(e + r), \quad \Lambda_{j+1}(r) = \log(e + \Lambda_j(r)). \quad (3)$$

Fix $r \geq 1$ with $\tau = \tau_r(A) > 0$ and put

$$t = \frac{r}{\tau}, \quad P(R) = \det(I + tR), \quad p_R(z) = \det(I + zR).$$

Since $\det(I + tR) \geq 1 + t \operatorname{tr} R$, we have $\operatorname{tr} R \leq P(R)/t$. The determinant potential controls exceptionally large residuals during the warm-start phases. Once the expected trace is a constant multiple of τ , the trace decrease alone supplies the accuracy phase. Every process is held at its terminal set after exhaustion; references to k steps mean at most k positive pivots.

2.1 Deterministic facts

Lemma 2.1 (Order and exhaustion). *Every positive pivot satisfies $0 \preceq R^{(i)} \preceq R$ and lowers the rank by one. Consequently every sequence of positive pivots satisfies $\operatorname{tr} R_k \geq \tau_k(A)$ and $\tau_r(R_k) \leq \tau_r(A)$, and exhausts A after $\operatorname{rank} A \leq N$ pivots.*

Proof. Put $u = R^{1/2}e_i/\sqrt{R_{ii}}$, a unit vector in the range of $R^{1/2}$. Then $R^{(i)} = R^{1/2}(I - uu^*)R^{1/2}$, which is PSD, is dominated by R , and has rank $\operatorname{rank} R - 1$. The matrix $A - R_k$ is PSD, dominated by A , and has rank at most k ; hence its trace is at most $\lambda_1 + \dots + \lambda_k$, which is the first inequality. The second follows from $R_k \preceq A$ and eigenvalue monotonicity. $\square$

Lemma 2.2 (Pivot products and exact path probabilities). *Let $w = (i_0, \dots, i_{k-1})$ be a sequence of positive pivots and $J_s = \{i_0, \dots, i_{s-1}\}$. Then*

$$\prod_{s \leq k} (A/J_s)_{i_s i_s} = \det A_{J_k J_k}, \quad \pi(w) = \frac{\det A_{J_k J_k}}{\prod_{s < k} \operatorname{tr}(A/J_s)},$$

where $\pi(w)$ is the probability that exact RPCholesky follows w .

Proof. The Schur determinant formula gives $\det A_{J_{s+1} J_{s+1}} = \det A_{J_s J_s} (A/J_s)_{i_s i_s}$. Induction gives the first identity, and the second is the product of the transition probabilities (1). $\square$

Lemma 2.3 (Exact drifts). *Let $r \geq 1$ and let $R \neq 0$ be a residual with $T = \text{tr } R$ and $\tau_r(R) \leq \tau$. Then*

$$\begin{aligned}\mathbb{E}_\pi[P(R') \mid R] &= \frac{p'_R(t)}{T}, \\ \mathbb{E}_\pi[\text{tr } R' \mid R] &= T - \frac{\text{tr } R^2}{T}.\end{aligned}\tag{4}$$

and

$$\begin{aligned}\frac{p'_R(t)}{T} &\leq 2e P(R)^{1-1/r}, \\ \frac{p'_R(t)}{P(R)} &\leq 2\tau, \\ \text{tr } R^2 &\geq \frac{(T - \tau)_+^2}{r}.\end{aligned}\tag{5}$$

Proof. By the matrix determinant lemma, $P(R^{(i)}) = P(R)(1 - t e_i^* R(I + tR)^{-1} R e_i / R_{ii})$. Averaging with weights R_{ii}/T gives $P(R)(1 - (t/T) \text{tr}[R^2(I + tR)^{-1}]) = (P(R)/T) \text{tr}[R(I + tR)^{-1}]$, and $\text{tr}[R(I + tR)^{-1}] = p'_R(t)/P(R)$. The trace identity follows from $\text{tr } R^{(i)} = T - \|R e_i\|^2 / R_{ii}$.

Let $x\tau$ be the sum of the r largest eigenvalues of R (padding by zeros) and $y\tau$ the rest, so $0 \leq y \leq 1$ and $T = (x + y)\tau$. AM–GM on the head and $\log(1 + u) \leq u$ on the tail give $P(R) \leq (1 + x)^r e^{xy}$. Concavity of $u \mapsto u/(1 + tu)$ on the head and $u/(1 + tu) \leq u$ on the tail give $p'_R(t)/P(R) = \sum_j \lambda_j / (1 + t\lambda_j) \leq \tau[x/(1 + x) + y]$. Hence

$$\frac{p'_R(t)}{T P(R)^{1-1/r}} \leq \frac{\tau[x/(1 + x) + y](1 + x)e^y}{(x + y)\tau} = e^y \left(1 + \frac{xy}{x + y}\right) \leq e(1 + y) \leq 2e.$$

The bound $\lambda/(1 + t\lambda) \leq 1/t$ on the r head eigenvalues and $\lambda/(1 + t\lambda) \leq \lambda$ on the tail give $p'_R(t)/P(R) \leq r/t + \tau = 2\tau$. Finally the head eigenvalues sum to at least $T - \tau$, so Cauchy–Schwarz gives $\text{tr } R^2 \geq (T - \tau)_+^2 / r$. $\square$

The trace excess is followed through the progress variable

$$F(u) = \frac{1}{u} - \log u, \quad u > 0,\tag{6}$$

which is decreasing and convex, with $F'(u) = -(1 + u)/u^2$.

Lemma 2.4 (Scalar completion). *Let $0 < a \leq 1$, $r \geq 1$, and $\phi_a(T) = T - a(T - \tau)_+^2 / (rT)$ for $T > 0$, $\phi_a(0) = 0$. Then ϕ_a is nondecreasing and concave on $[0, \infty)$. If nonnegative numbers f_k satisfy $f_{k+1} \leq \phi_a(f_k)$, then as long as $u_k = f_k/\tau - 1 > 0$,*

$$F(u_{k+1}) \geq F(u_k) + \frac{a}{r},$$

where $F(u_{k+1}) = +\infty$ if $u_{k+1} \leq 0$. Consequently, if $f_{k_0} \leq (1 + u)\tau$ with $u > \varepsilon$, then $f_k \leq (1 + \varepsilon)\tau$ for $k \geq k_0 + \lceil (r/a)(F(\varepsilon) - F(u)) \rceil$.

Proof. On $[0, \tau]$, $\phi_a(T) = T$. For $T > \tau$, $\phi'_a(T) = 1 - (a/r)(1 - \tau^2/T^2) \geq 1 - a/r \geq 0$ and $\phi''_a(T) = -2a\tau^2/(rT^3) < 0$; the one-sided derivatives agree at τ . This proves monotonicity and

concavity. Normalizing, $u_{k+1} \leq u_k - au_k^2/[r(1+u_k)]$, and the right side is positive. Since F is decreasing and convex,

$$F(u_{k+1}) \geq F\left(u_k - \frac{au_k^2}{r(1+u_k)}\right) \geq F(u_k) - F'(u_k)\frac{au_k^2}{r(1+u_k)} = F(u_k) + \frac{a}{r}.$$

Since $\phi_a(T) \leq T$, the sequence f_k is nonincreasing once it is below $(1+\varepsilon)\tau$, and the count follows. $\square$

2.2 The imported initial moment

We use the following result as a black box. Put $b_r = \log(1 + 2\log(r!)/r)$ and $a_r = r^r/r!$. For $r \geq 2$ and $0 < \beta < 1$, define the optimizing exponent and its logarithmic cost by

$$\alpha_{r,\beta} = \min\left\{\beta, \frac{r-1}{rb_r}\right\},$$

$$B_{r,\beta} = rb_r\alpha_{r,\beta} - (r-1)\log \alpha_{r,\beta}.$$

Let $\varphi = (1 + \sqrt{5})/2$ and $c_v = 1 + 2\log \varphi + \varphi^{-1}$. The imported moment scale is

$$H_{r,\beta}^{(4)} = B_{r,\beta} + \log a_r + \beta r \log 2 + (1 - \beta)c_v r. \quad (7)$$

Imported theorem 2.5 ([8, Proposition 7.2 and Theorem 6.3]). *Let $r \geq 2$ and $0 < \beta < 1$. For every finite PSD matrix A with $\tau_r(A) > 0$, exact RPCholesky satisfies*

$$\mathbb{E}_\pi P(R_r)^{1-\beta} \leq \exp(H_{r,\beta}^{(4)}). \quad (8)$$

Moreover, for every PSD A with $\text{rank } A \geq r$ (including $\text{rank } A = r$), let μ_r be the law of the unordered set of the first r pivots, $\nu_r(S) = \det A_{SS}/e_r(A)$ the volume-sampling law, and $L_A = \mu_r/\nu_r$. Then

$$\mathbb{E}_{\nu_r} L_A^{1/\beta} \leq e^{B_{r,\beta}/\beta} 2^r a_r^{1/\beta}. \quad (9)$$

For fixed β , $B_{r,\beta} = O(r \log \log \log r)$ and $H_{r,\beta}^{(4)}/r = O(\log \log \log r)$.

The same paper converts (8) into the count $K_4(r, \varepsilon) = O(r/\varepsilon + r\Lambda_4(r))$ for exact RPCholesky; its conversion is the case $a = 1$ of Proposition 3.1 below, and the resulting counts at $\varepsilon = 1$ are 71, 722, 7243, 72539 and 726105 for $r = 10, 10^2, \dots, 10^5$, with β optimized for each rank. We have recomputed these values.

3 A functional robust conversion

Throughout this section $(R_k)_{k \geq 0}$ is a sequence of residuals of A produced by positive pivots and exact Schur updates, adapted to a filtration $(\mathcal{F}_k)$. The filtration contains the pivots chosen so far and may contain arbitrary external randomness, such as a sketching matrix. No independence between steps is assumed.

3.1 Statement

Fix $r \geq 2$, $\tau = \tau_r(A) > 0$, $0 < \varepsilon \leq 1$, $0 < a \leq 1$, $0 < \theta \leq 1$ and a real number H . Put $\zeta = 1 - 1/r$. The four phase lengths are r , g_a , s_a and n_a . We define them in their order of use. Cooling reduces the moment exponent:

$$c_a = \log(2e/a), \quad g_a = \left\lceil \frac{\log \max\{\theta^{-1}, [H/(r\theta) - c_a]/\log 5, 1\}}{-\log(1 - 1/r)} \right\rceil. \quad (10)$$

Its remaining first-moment scale is

$$\rho_a = \zeta^{g_a}, \quad D_a = (1 - \rho_a)rc_a + \rho_a H/\theta. \quad (11)$$

The stopping phase reaches a constant expected trace excess:

$$s_a = \lceil D_a/\log 5 \rceil, \quad u_a = 10/a - 1 + 1/r. \quad (12)$$

Finally, the scalar progress function fixes the accuracy phase:

$$n_a = \left\lceil \frac{r}{a} (F(\varepsilon) - F(u_a)) \right\rceil, \quad (13)$$

$$\mathcal{K}_a(r, \varepsilon; \theta, H) = r + g_a + s_a + n_a.$$

At $a = 1$ these are exactly the formulas of the exact-probability conversion in [8].

Proposition 3.1 (Functional robust conversion). *Assume*

$$(I) \quad \mathbb{E}P(R_r)^\theta \leq e^H;$$

$$(U) \quad \text{for every } k \geq r, \text{ on } \{R_k \neq 0\}, \mathbb{E}[P(R_{k+1}) \mid \mathcal{F}_k] \leq a^{-1}p'_{R_k}(t)/\text{tr } R_k;$$

$$(L) \quad \text{for every } k \geq r, \text{ on } \{R_k \neq 0\}, \mathbb{E}[\text{tr } R_{k+1} \mid \mathcal{F}_k] \leq \text{tr } R_k - a \text{tr } R_k^2 / \text{tr } R_k.$$

Then $\mathbb{E} \text{tr } R_k \leq (1 + \varepsilon)\tau$ for every $k \geq \min\{N, \mathcal{K}_a(r, \varepsilon; \theta, H)\}$.

By (4), the right sides of (U) and (L) are a^{-1} times the exact conditional mean of $P(R')$, and the exact mean trace decrease multiplied by a . The hypotheses therefore compare only two functionals of the pivot law with their exact values: the determinant functional $i \mapsto P(R^{(i)})$ from above, and the trace decrease $i \mapsto \|Re_i\|^2/R_{ii}$ from below.

3.2 Proof, phase by phase

The residual is zero from some time on only if it is exhausted, and a zero residual stays zero; on $\{R_k = 0\}$ all the bounds below hold trivially because $P(0) = 1$ and $\text{tr } 0 = 0$. After rank $A \leq N$ positive pivots the residual is zero, which gives the cap at N . Pathwise $\text{tr } R_k$ is nonincreasing (Lemma 2.1), so it suffices to prove the bound at $k = \mathcal{K}_a$.

Phase 1: initialization (r pivots). This phase is hypothesis (I).

Phase 2: cooling (g_a pivots). By (U) and the first inequality in (5), for $k \geq r$,

$$\mathbb{E}[P(R_{k+1}) \mid \mathcal{F}_k] \leq \frac{2e}{a} P(R_k)^\zeta.$$

We claim that for $j \geq 0$, $\mathbb{E}[P(R_{r+j}) \mid \mathcal{F}_r] \leq (2e/a)^{r(1-\zeta^j)} P(R_r)^{\zeta^j}$. For $j = 0$ this is trivial. If it holds for j , then by the tower property and conditional Jensen for the concave map $x \mapsto x^\zeta$,

$$\mathbb{E}[P(R_{r+j+1}) \mid \mathcal{F}_r] \leq \frac{2e}{a} \mathbb{E}[P(R_{r+j})^\zeta \mid \mathcal{F}_r] \leq \frac{2e}{a} \left((2e/a)^{r(1-\zeta^j)} P(R_r)^{\zeta^j} \right)^\zeta,$$

and $1 + r\zeta(1 - \zeta^j) = r(1 - \zeta^{j+1})$ because $r(1 - \zeta) = 1$. Take $j = g_a$. The first entry of the maximum in (10) gives $\rho_a \leq \theta$, so Jensen and (I) give

$$\log \mathbb{E}P(R_{r+g_a}) \leq (1 - \rho_a)rc_a + \log \mathbb{E}P(R_r)^{\rho_a} \leq (1 - \rho_a)rc_a + \frac{\rho_a}{\theta} \log \mathbb{E}P(R_r)^\theta \leq D_a.$$

The second entry gives $\rho_a[H/(r\theta) - c_a] \leq \log 5$, which is $D_a \leq r(c_a + \log 5)$.

Phase 3: stopping (s_a pivots). Let $k \geq r + g_a$ and suppose $\text{tr } R_k > 10\tau/a$. By (U) and the second inequality in (5),

$$\mathbb{E}[P(R_{k+1}) \mid \mathcal{F}_k] \leq \frac{2\tau}{a \text{tr } R_k} P(R_k) \leq \frac{1}{5} P(R_k).$$

Put $Z_k = P(R_k) \mathbf{1}\{\text{tr } R_k > 10\tau/a\}$. Since the trace is pathwise nonincreasing, $\mathbf{1}\{\text{tr } R_{k+1} > 10\tau/a\} \leq \mathbf{1}\{\text{tr } R_k > 10\tau/a\}$, and therefore $\mathbb{E}[Z_{k+1} \mid \mathcal{F}_k] \leq Z_k/5$. After s_a steps $\mathbb{E}Z_{r+g_a+s_a} \leq 5^{-s_a} e^{D_a} \leq 1$. Since $\text{tr } R \leq P(R)/t$,

$$\mathbb{E} \text{tr } R_{r+g_a+s_a} \leq \frac{10\tau}{a} + \frac{\mathbb{E}Z_{r+g_a+s_a}}{t} \leq \left(\frac{10}{a} + \frac{1}{r} \right) \tau = (1 + u_a)\tau.$$

Phase 4: accuracy (n_a pivots). By (L), Lemma 2.1 ($\tau_r(R_k) \leq \tau$) and the last inequality in (5), $\mathbb{E}[\text{tr } R_{k+1} \mid \mathcal{F}_k] \leq \phi_a(\text{tr } R_k)$ with ϕ_a as in Lemma 2.4. Since ϕ_a is concave and nondecreasing, Jensen gives $f_{k+1} \leq \phi_a(f_k)$ for the means $f_k = \mathbb{E} \text{tr } R_k$. Lemma 2.4, started from $u \leq u_a$, gives $f_k \leq (1 + \varepsilon)\tau$ after n_a further pivots. This proves the proposition. $\square$

Remark 3.2 (Which side is used where). Phases 2 and 3 use only (U), an upper comparison. Phase 4 uses only (L), a lower comparison. Phase 1 is a moment hypothesis; under a per-coordinate bracket it is obtained from the upper side through the path likelihood. This separation is the source of the trichotomy in Section 4.

3.3 Two-sided brackets

Corollary 3.3 (Two-sided robust count, recalled from [8, Proposition 8.1]). *Let $r \geq 2$ and suppose that, conditionally on the complete past, the pivot law satisfies*

$$a\pi_i \leq q_i \leq a^{-1}\pi_i \quad \text{for all } i, \quad 0 < a \leq 1. \quad (14)$$

If exact RPCholesky satisfies $\mathbb{E}_\pi P(R_r)^\theta \leq e^H$, then $\mathbb{E} \text{tr } R_k \leq (1 + \varepsilon)\tau$ for all $k \geq \min\{N, K_a\}$, where

$$K_a = r + g_a + s_a + n_a = \mathcal{K}_a(r, \varepsilon; \theta, H + r \log(1/a)).$$

With Imported Theorem 2.5, $\theta = 1 - \beta$ and $H = H_{r,\beta}^{(4)}$, for fixed $0 < \beta < 1$,

$$K_a = O\left(\frac{r}{a\varepsilon} + \frac{r}{a} \log \frac{e}{a} + r\Lambda_4(r)\right), \quad (15)$$

with a constant independent of a , ε and r .

Proof. (I). By induction on the steps, the probability that the process follows a pivot sequence w of length r is at most $a^{-r}\pi(w)$: each conditional factor $q(i_s | \mathcal{F}_s)$ is at most $a^{-1}\pi(i_s | J_s)$, and $\pi(\cdot | J_s)$ depends only on the set J_s . Since $P(R_r)^\theta \geq 0$ is a function of the sequence, $\mathbb{E}_q P(R_r)^\theta \leq a^{-r} \mathbb{E}_\pi P(R_r)^\theta \leq e^{H+r \log(1/a)}$.

(U). $P(R^{(i)}) \geq 0$, so $\sum_i q_i P(R^{(i)}) \leq a^{-1} \sum_i \pi_i P(R^{(i)}) = a^{-1} p'_R(t) / \text{tr } R$ by (4).

(L). The decrease $\|Re_i\|^2/R_{ii}$ is nonnegative, so $\sum_i q_i \|Re_i\|^2/R_{ii} \geq a \sum_i \pi_i \|Re_i\|^2/R_{ii} = a \text{tr } R^2 / \text{tr } R$.

Proposition 3.1 gives the count. For the asymptotics, $u_a \leq 10/a$ and $F(u_a) \geq -\log u_a$, so $n_a \leq 1 + (r/a)(\varepsilon^{-1} + \log \varepsilon^{-1} + \log(10/a))$. Next $D_a \leq r(c_a + \log 5)$ gives $s_a = O(r \log(e/a))$. Finally $-1/\log(1 - 1/r) \leq r$ and $H_{r,\beta}^{(4)}/r = O(\log \log \log r)$ give

$$g_a \leq 1 + r \log \max \left\{ \frac{1}{1-\beta}, C(1 + \log \log \log(e^e + r) + \log(1/a)) \right\} = O(r \Lambda_4(r) + r \log(e/a)),$$

using $\log(x+y) \leq \log x + \log(1+y)$ for $x \geq 1, y \geq 0$. Since $\log \varepsilon^{-1} \leq \varepsilon^{-1}$ and $r \log(e/a) \leq (r/a) \log(e/a)$, the bound (15) follows. $\square$

Remark 3.4 (Rank one and zero tails). For $r = 1$ the upper side applied to the nonnegative function $i \mapsto \text{tr } R^{(i)}$ at the first step gives $\mathbb{E}_q \text{tr } R_1 \leq a^{-1} \mathbb{E}_\pi \text{tr } R_1 \leq 2\tau_1(A)/a$, because $\mathbb{E}_\pi \text{tr } R_1 \leq (\lambda_1 + \tau) - \lambda_1^2/(\lambda_1 + \tau) \leq 2\tau$. Lemma 2.4 with $r = 1$ then shows that $1 + \lceil a^{-1}[F(\varepsilon) - F(2/a - 1)] \rceil$ pivots suffice. If $\tau_r(A) = 0$, positive pivots exhaust A after at most r steps. Any count valid for rank 2 is also valid for the rank-one target, since $\tau_2(A) \leq \tau_1(A)$.

At $a = 1$ the corollary is the four-logarithm theorem of [8]. Table 2 in Section 8 evaluates K_a : at $r = 1000$ and $\varepsilon = 1$ the count grows from 7243 at $a = 1$ to 12393 at $a = 1/2$, and at $\varepsilon = 0.1$ from 18545 to 34998. The dominant change is the factor $1/a$ in the accuracy phase.

4 The robustness trichotomy

Remark 3.2 says which side of the bracket each phase uses. This section shows that the division is sharp: a lower bracket alone keeps the final accuracy phase, and an upper bracket alone keeps the first three phases but not the accuracy.

4.1 Lower brackets

Theorem 4.1 (Lower brackets keep the scalar completion). *Let $r \geq 1$, $\tau = \tau_r(A) > 0$, $0 < a \leq 1$, and suppose the pivot law satisfies $q_i \geq a\pi_i$ for all i , conditionally on the complete past. If $\mathbb{E} \text{tr } R_{k_0} \leq (1+u)\tau$ with $u > \varepsilon$, then $\mathbb{E} \text{tr } R_k \leq (1+\varepsilon)\tau$ for all $k \geq k_0 + \lceil (r/a)(F(\varepsilon) - F(u)) \rceil$. In particular*

$$k \geq \left\lceil \frac{r}{a} \left(\frac{1}{\varepsilon} + \log \frac{1}{\varepsilon} + \log \frac{\text{tr } A}{\tau} \right) \right\rceil$$

pivots suffice from $R_0 = A$.

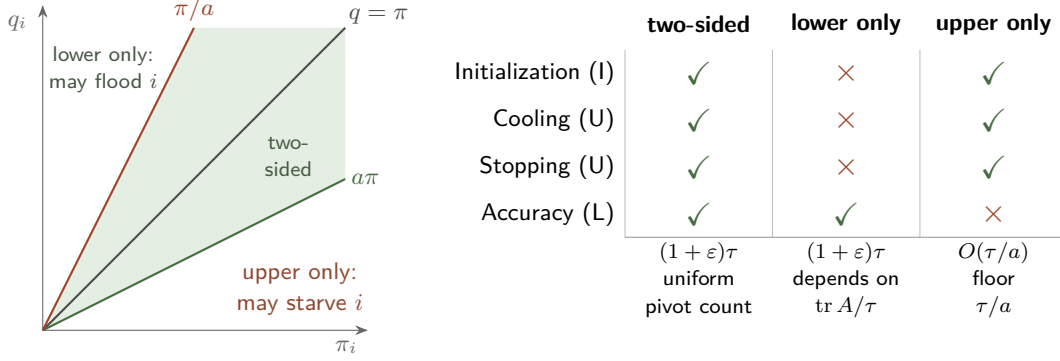


Figure 2: The robustness trichotomy. Left: the admissible region for a pivot probability q_i given π_i . Right: which phases of the conversion survive, and the resulting guarantee. The lower-only count contains $\log(\text{tr } A/\tau)$ and starts from the input (or from any warm start); the upper-only floor is Theorem 4.2(b).

Proof. The proof of (L) in Corollary 3.3 uses only the lower side, so (L) holds at every step. Phase 4 of the proof of Proposition 3.1 applies verbatim from time k_0 . From $R_0 = A$ we have $u = \text{tr } A/\tau - 1$ and $F(u) > -\log u > -\log(\text{tr } A/\tau)$, while $F(\varepsilon) = \varepsilon^{-1} + \log \varepsilon^{-1}$. $\square$

Thus a lower bracket never produces an accuracy floor: every relative accuracy is reached, at a count that depends on the starting excess only logarithmically. What is lost is the uniformity in $\text{tr } A/\tau$: the determinant phases that remove this dependence need an upper comparison.

4.2 Upper brackets

Theorem 4.2 (Upper brackets: constant factor, and a floor). *Let $0 < a \leq 1$.*

- (a) *Let $r \geq 2$ and $\tau = \tau_r(A) > 0$. If $q_i \leq \pi_i/a$ for all i , conditionally on the complete past, and exact RPCholesky satisfies $\mathbb{E}_\pi P(R_r)^\theta \leq e^H$, then $\mathbb{E} \text{tr } R_k \leq (10/a + 1/r)\tau$ for all $k \geq r + g_a + s_a$, with g_a, s_a computed from $(\theta, H + r \log(1/a))$.*
- (b) *For every $r \geq 1$, every budget $k \geq 0$ and every $\delta > 0$ there are m and a memoryless pivot rule with $q \leq \pi/a$ such that, on*

$$A = \text{diag}(\Theta I_r, I_m), \quad \Theta = \frac{(m-k)(1-a)}{ar},$$

the residual after k pivots is deterministic and

$$\frac{\text{tr } R_k}{\tau_r(A)} \geq \frac{1 - k/m}{a} \geq \frac{1}{a} - \delta.$$

Proof. (a) The proof of (I) and (U) in Corollary 3.3 uses only the upper side. Phases 1–3 of Proposition 3.1 then give the stated bound, and the trace is nonincreasing afterwards.

(b) Take $m > k + r$ and $m \geq k/(a\delta)$. This ensures a positive rank- r tail even when $a = 1$, for which $\Theta = 0$. Label the first r coordinates *heads* and the last m *tails*. Every residual reached by positive pivots on a diagonal matrix is the same diagonal matrix with the pivoted coordinates removed; a state is described by the numbers h of heads and b of tails that remain. The rule is as follows: if $b \geq 1$ and $h\Theta \leq b(1-a)/a$, choose a remaining tail uniformly; otherwise use

exact RPCholesky. In the first case, a remaining tail i has $q_i = 1/b$ and $\pi_i = 1/(h\Theta + b)$, and $q_i \leq \pi_i/a$ is equivalent to $a(h\Theta + b) \leq b$, which is the case condition; heads have $q_i = 0$. So the rule satisfies $q \leq \pi/a$ at every state.

Start from $h = r$, $b = m$. Before the $(s+1)$ st pivot, $s < k$, we have $b = m - s \geq m - k + 1$ and $r\Theta = (m - k)(1 - a)/a \leq b(1 - a)/a$, so the rule picks a tail. After k pivots the residual is $\text{diag}(\Theta I_r, I_{m-k})$ and

$$\text{tr } R_k = r\Theta + m - k = \frac{(m - k)(1 - a)}{a} + (m - k) = \frac{m - k}{a}.$$

The eigenvalues of A are Θ (r times) and 1 (m times), so $\tau_r(A) \leq m$ in either order. Hence $\text{tr } R_k / \tau_r(A) \geq (1 - k/m)/a \geq 1/a - \delta$. $\square$

Remark 4.3. In the construction the heads are starved: $q = 0 < a\pi$ on them. Any lower bracket rules this out, and Theorem 4.1 then forces the error to $(1 + \varepsilon)\tau$. Conversely, the construction does not contradict part (a): when $\Theta \geq 1$, $\tau_r(A) = m$ and $\text{tr } R_k \leq \text{tr } A \leq m/a$ for every k , so the floor $1/a$ is also the largest possible ratio on this family. The floor holds at every finite budget, but the input depends on the budget through m . Section 8 evaluates, by an exact dynamic program, the four laws on this family: exact, two-sided, lower-only and upper-only adversaries.

Proof of Theorem 1.1. Part (i) is Corollary 3.3 for $r \geq 2$ and Remark 3.4 for $r = 1$. Part (ii) is Theorem 4.1. Part (iii) is Theorem 4.2. $\square$

5 Sketched RPCholesky

5.1 The algorithm and two exact identities

Let $F \in \mathbb{C}^{D \times N}$ and $A = F^*F$. Let $\Pi \in \mathbb{C}^{m \times D}$ be a random matrix, independent of the pivot randomness, and put $B = F^*\Pi^*\Pi F$. *Sketched RPCholesky* runs exact RPCholesky on B : at a pivot set J with $\text{tr}(B/J) > 0$ it chooses i with probability

$$q_\Pi(i \mid J) = \frac{(B/J)_{ii}}{\text{tr}(B/J)},$$

and it stops early if $\text{tr}(B/J) = 0$. A stopped chain keeps its final set at all later times; an exhausted reference or hybrid chain uses the same convention. After K steps it returns the set J ; the error is measured on A by $\text{tr}(A/J)$. It may also return the same-sketch reconstruction

$$X_B = (\Pi F_J)^+ \Pi F, \quad F \approx F_J X_B.$$

We write $R_A(J) = A/J$ and $R_B(J) = B/J$.

For a set J such that F_J has full column rank, let $Q_J \in \mathbb{C}^{D \times |J|}$ have orthonormal columns spanning the range of F_J , and put

$$E_J = (I - Q_J Q_J^*)F, \quad G_J = Q_J^* \Pi^* \Pi Q_J.$$

Then $R_A(J) = E_J^* E_J$. When G_J is invertible, let $P_J = \Pi Q_J G_J^{-1} Q_J^* \Pi^*$ be the orthogonal projector onto the range of ΠQ_J , which is the range of ΠF_J .

Lemma 5.1 (Sketched residual). *If F_J and ΠF_J have full column rank, then*

$$R_B(J) = (\Pi E_J)^*(I - P_J)(\Pi E_J).$$

In particular $(R_B(J))_{ii} = \|(I - P_J)\Pi E_J e_i\|^2$, and $(R_A(J))_{ii} = 0$ implies $(R_B(J))_{ii} = 0$. Along every path of sketched RPCholesky, F_J and ΠF_J keep full column rank, and every pivot is positive for both A and B .

Proof. The Schur complement of a Gram matrix is the Gram matrix of the projected residual: $B/J = (\Pi F)^*(I - P_J)(\Pi F)$. Write $F = F_J X + E_J$ with $X = (F_J^* F_J)^{-1} F_J^* F$; then $\Pi F = \Pi F_J X + \Pi E_J$ and $(I - P_J)\Pi F_J = 0$. If $(R_A)_{ii} = 0$ then $E_J e_i = 0$. A pivot i is chosen only if $(R_B)_{ii} > 0$, which means $\Pi F e_i \notin \text{range } \Pi F_J$; hence $\Pi F_{J \cup \{i\}}$, and therefore $F_{J \cup \{i\}}$, has full column rank, and $(R_A)_{ii} > 0$. Induction on the steps completes the proof. $\square$

Lemma 5.2 (Exact path-likelihood identity). *Let $w = (i_0, \dots, i_{t-1})$ be a pivot sequence with positive pivots for B , and let $J_s = \{i_0, \dots, i_{s-1}\}$. Let $q_\Pi(w)$ and $p(w)$ be the probabilities that sketched RPCholesky and exact RPCholesky on A follow w . Then*

$$\frac{q_\Pi(w)}{p(w)} = \det(Q_{J_t}^* \Pi^* \Pi Q_{J_t}) \prod_{s < t} \frac{\text{tr } R_A(J_s)}{\text{tr } R_B(J_s)}.$$

Proof. Lemma 2.2 for A and for B gives $q_\Pi(w) = \det B_{J_t J_t} / \prod_s \text{tr } R_B(J_s)$ and $p(w) = \det A_{J_t J_t} / \prod_s \text{tr } R_A(J_s)$. Write $F_{J_t} = Q_{J_t} C$ with C invertible. Then $A_{J_t J_t} = C^* C$ and $B_{J_t J_t} = C^* G_{J_t} C$, so the ratio of determinants is $\det G_{J_t}$. $\square$

The identity shows that sketched pivoting reads only two things: the distortion of the path subspace $\text{range } F_{J_t}$, and the ratio of residual traces along the path. Both are properties of subspaces of dimension at most $t + 1$, and the reference law p does not depend on Π .

5.2 The moment property

For an isometry U put $Z_U = U^* \Pi^* \Pi U - I$.

Definition 5.3. The random matrix Π has the *moment property* $\text{MP}(d, \ell, \varepsilon_0)$ if $\mathbb{E} \text{tr } Z_U^{2\ell} \leq d \varepsilon_0^{2\ell}$ for every isometry $U \in \mathbb{C}^{D \times d'}$ with $d' \leq d$.

Remark 5.4. (i) If $D \geq d$ and the bound holds for every isometry with exactly d columns, it holds for fewer columns: extend U to an isometry U' with d columns; then Z_U is a compression of $Z_{U'}$, and Schatten norms do not increase under compression, so $\text{tr } Z_U^{2\ell} \leq \text{tr } Z_{U'}^{2\ell}$. If $D < d$, pad F with zero rows; for sketches with independent columns, such as SparseStack, this does not change ΠF in law. (ii) A real sketch applied to complex data: if U is a complex isometry with d columns, let W be a real isometry whose range is the real span of the real and imaginary parts of the columns of U ; it has at most $2d$ columns and $U = WC$ with $C^* C = I$. Then $Z_U = C^* Z_W C$, so $\text{MP}(2d, \ell, \varepsilon_0)$ for real isometries gives the complex bound $\mathbb{E} \text{tr } Z_U^{2\ell} \leq 2d \varepsilon_0^{2\ell}$. This has twice the prefactor in Definition 5.3; it is not literally $\text{MP}(d, \ell, \varepsilon_0)$. The bad-state bound is therefore doubled, and the first-failure bound is multiplied by $\sqrt{2}$. Increasing the prescribed ℓ by one absorbs that factor. (iii) Markov's inequality and $\|Z\|^{2\ell} \leq \text{tr } Z^{2\ell}$ give $\mathbb{P}\{\|Z_U\| > \eta\} \leq d(\varepsilon_0/\eta)^{2\ell}$, the usual subspace embedding statement.

5.3 The good event on path-local subspaces

Fix $K \geq 1$, $0 < \eta \leq 1/4$, a set J with $|J| \leq K$ such that F_J has full column rank, and suppose $R_A(J) \neq 0$. Let Σ_J be the set of i with $(R_A(J))_{ii} > 0$. For $i \in \Sigma_J$ put

$$u_i = \frac{E_J e_i}{\|E_J e_i\|}, \quad U_i = [Q_J \quad u_i], \quad Z_i = U_i^* \Pi^* \Pi U_i - I.$$

Since $Q_J^* u_i = 0$, U_i is an isometry with at most $K + 1$ columns. Define three probability vectors on Σ_J :

$$\begin{aligned} \omega_i^{(1)} &= \frac{(R_A)_{ii}}{\text{tr } R_A}, \\ \omega_i^{(2)} &= \frac{\|R_A e_i\|^2}{\text{tr } R_A^2}, \\ \omega_i^{(3)} &= \frac{(R_A)_{ii} P(R_A^{(i)})}{\sum_j (R_A)_{jj} P(R_A^{(j)})}. \end{aligned}$$

where $R_A = R_A(J)$, $R_A^{(i)} = A/(J \cup \{i\})$, and P is the determinant potential of Section 2. The first is the exact pivot law, the second weights the exact trace decrease, and the third weights the exact determinant drift. For $c = 1, 2, 3$ put

$$\Phi_c(J) = \sum_{i \in \Sigma_J} \omega_i^{(c)} \|Z_i\|^4.$$

The *good event* at J is

$$\mathcal{G}(J) = \{\|G_J - I\| \leq \eta\} \cap \{\Phi_c(J) \leq \eta^4/4 \text{ for } c = 1, 2, 3\}. \quad (16)$$

If $R_A(J) = 0$ we set $\mathcal{G}(J)$ to be the sure event. The frames U_i and the weights $\omega_i^{(c)}$ are determined by F , J , r and τ ; they do not depend on Π . A state is *bad* if its good event fails.

Lemma 5.5 (Bad states are rare). *Let Π have $\text{MP}(K + 1, \ell, \varepsilon_0)$ with $\ell \geq 2$ and $\varepsilon_0 \leq \eta/2$. For every fixed J with $|J| \leq K$,*

$$\mathbb{P}\{\mathcal{G}(J)^c\} \leq (K + 1) \left[\left(\frac{\varepsilon_0}{\eta} \right)^{2\ell} + 3 \left(\frac{2\varepsilon_0^2}{\eta^2} \right)^\ell \right] \leq 4(K + 1)2^{-\ell} =: \delta_1.$$

Proof. Remark 5.4(iii) bounds the first event. For the others, $\|Z_i\|^{2\ell} \leq \text{tr } Z_i^{2\ell}$ gives $\mathbb{E}\|Z_i\|^{2\ell} \leq (K + 1)\varepsilon_0^{2\ell}$. Since $\ell/2 \geq 1$, Minkowski's inequality in $L^{\ell/2}$ applied to the nonnegative variables $\omega_i^{(c)}\|Z_i\|^4$ gives

$$(\mathbb{E}\Phi_c^{\ell/2})^{2/\ell} \leq \sum_i \omega_i^{(c)} (\mathbb{E}\|Z_i\|^{2\ell})^{2/\ell} \leq ((K + 1)\varepsilon_0^{2\ell})^{2/\ell}.$$

Markov's inequality gives $\mathbb{P}\{\Phi_c > \eta^4/4\} \leq (K + 1)\varepsilon_0^{2\ell}(4/\eta^4)^{\ell/2} = (K + 1)(2\varepsilon_0^2/\eta^2)^\ell$. With $\varepsilon_0 \leq \eta/2$ the four terms are at most $(K + 1)(4^{-\ell} + 3 \cdot 2^{-\ell}) \leq 4(K + 1)2^{-\ell}$. $\square$

No union bound over the N coordinates occurs: each of the four events is a statement about one weighted average over coordinates, and Minkowski's inequality averages the per-coordinate moment bounds.

5.4 What a good state guarantees

Put

$$\gamma_\eta = \frac{1 - \eta - \eta^2}{1 + \eta}, \quad \chi_\eta^2 = \frac{(\eta/\sqrt{2} + \eta^2/(2(1 - \eta)))^2}{(1 - \eta - \eta^2)^2}. \quad (17)$$

For $\eta = 0.1$ and $\eta = 1/4$ these are $\gamma_\eta = 0.809$, $\chi_\eta^2 = 0.0073$ and $\gamma_\eta = 0.55$, $\chi_\eta^2 = 0.101$. As $\eta \rightarrow 0$, $1 - \gamma_\eta \sim 2\eta$ and $\chi_\eta^2 \sim \eta^2/2$.

Lemma 5.6 (Consequences of a good state). *Let $\mathcal{G}(J)$ hold, $R_A(J) \neq 0$, and put $y_i = (R_B(J))_{ii}/(R_A(J))_{ii}$ for $i \in \Sigma_J$. Then*

- (a) $G_J \succeq (1 - \eta)I$, so ΠF_J has full column rank;
- (b) $1 - \eta - \eta^2 \leq \sum_i \omega_i^{(c)} y_i \leq 1 + \eta$ for $c = 1, 2, 3$; in particular
$$(1 - \eta - \eta^2) \operatorname{tr} R_A(J) \leq \operatorname{tr} R_B(J) \leq (1 + \eta) \operatorname{tr} R_A(J); \quad (18)$$
- (c) $\sum_i \omega_i^{(1)} (y_i - 1)^2 \leq (\eta/\sqrt{2} + \eta^2/(2(1 - \eta)))^2$;
- (d) $\|Q_J^* \Pi^* \Pi E_J\|_F^2 \leq (\eta^2/2) \operatorname{tr} R_A(J)$.

Proof. Part (a) is the first event in (16). To prove the remaining claims, write $Q = Q_J$, $G = G_J$ and $P = P_J$. The residual identity separates two sources of distortion:

$$\begin{aligned} y_i &= \|(I - P)\Pi u_i\|^2 = 1 + z_i - p_i, \\ z_i &= \|\Pi u_i\|^2 - 1, \\ p_i &= \|P\Pi u_i\|^2 = c_i^* G^{-1} c_i, \quad c_i = Q^* \Pi^* \Pi u_i. \end{aligned}$$

Here z_i is the norm distortion and p_i is the loss from projection onto the sketched selected span. Because $Q^* u_i = 0$, they are controlled by the diagonal and off-diagonal blocks of Z_i :

$$|z_i| \leq \|Z_i\|, \quad \|c_i\| \leq \|Z_i\|, \quad 0 \leq p_i \leq \frac{\|c_i\|^2}{1 - \eta}.$$

Weighted first moments, for (b). Let ω be any of the three probability vectors. Its fourth-moment test gives, by Jensen's inequality,

$$\begin{aligned} \sum_i \omega_i |z_i| &\leq \left(\sum_i \omega_i z_i^4 \right)^{1/4} \leq \frac{\eta}{\sqrt{2}}, \\ \sum_i \omega_i p_i &\leq \frac{(\sum_i \omega_i \|c_i\|^4)^{1/2}}{1 - \eta} \leq \frac{\eta^2}{2(1 - \eta)}. \end{aligned}$$

Combining these estimates with $y_i = 1 + z_i - p_i$ shows that

$$\sum_i \omega_i y_i \in \left[1 - \frac{\eta}{\sqrt{2}} - \frac{\eta^2}{2(1 - \eta)}, 1 + \frac{\eta}{\sqrt{2}} \right].$$

For $\eta \leq 1/4$ this interval lies in $[1 - \eta - \eta^2, 1 + \eta]$. The trace comparison is its specialization to $\omega^{(1)}$, since $\operatorname{tr} R_B = \sum_i (R_A)_{ii} y_i$.

Weighted second moments, for (c). For $\omega = \omega^{(1)}$, the same fourth-moment test gives

$$\sum_i \omega_i z_i^2 \leq \frac{\eta^2}{2}, \quad \sum_i \omega_i p_i^2 \leq \frac{\eta^4}{4(1 - \eta)^2}.$$

Minkowski's inequality in $L^2(\omega)$ therefore yields

$$\left(\sum_i \omega_i (y_i - 1)^2 \right)^{1/2} \leq \left(\sum_i \omega_i z_i^2 \right)^{1/2} + \left(\sum_i \omega_i p_i^2 \right)^{1/2} \leq \frac{\eta}{\sqrt{2}} + \frac{\eta^2}{2(1-\eta)}.$$

The reconstruction cross term, for (d). Each column of E_J has squared norm $(R_A)_{ii}$. Thus

$$\begin{aligned} \|Q^* \Pi^* \Pi E_J\|_F^2 &= \text{tr } R_A \sum_i \omega_i^{(1)} \|c_i\|^2 \\ &\leq \text{tr } R_A \left(\sum_i \omega_i^{(1)} \|c_i\|^4 \right)^{1/2} \leq \frac{\eta^2}{2} \text{tr } R_A. \end{aligned}$$

□

Lemma 5.7 (The sketched kernel at a good state). *Let $\mathcal{G}(J)$ hold and $R_A(J) \neq 0$; write $R = R_A(J)$, $\pi = \pi(R)$, $q = q_\Pi(\cdot | J)$. Then*

- (a) (upper functional) $\sum_i q_i P(R^{(i)}) \leq \gamma_\eta^{-1} \sum_i \pi_i P(R^{(i)}) = \gamma_\eta^{-1} p'_R(t) / \text{tr } R$;
- (b) (lower functional) $\sum_i q_i \|Re_i\|^2 / R_{ii} \geq \gamma_\eta \text{tr } R^2 / \text{tr } R$;
- (c) (per-step χ^2) $\sum_i q_i^2 / \pi_i \leq 1 + \chi_\eta^2$.

Proof. By Lemma 5.1, q is supported in Σ_J and $q_i = \pi_i y_i / \bar{y}$ with $\bar{y} = \sum_i \omega_i^{(1)} y_i = \text{tr } R_B / \text{tr } R$. For (a), $\pi_i P(R^{(i)})$ is proportional to $\omega_i^{(3)}$, so

$$\sum_i q_i P(R^{(i)}) = \left(\sum_i \pi_i P(R^{(i)}) \right) \frac{\sum_i \omega_i^{(3)} y_i}{\bar{y}} \leq \frac{1 + \eta}{1 - \eta - \eta^2} \sum_i \pi_i P(R^{(i)})$$

by Lemma 5.6(b), and (4) identifies the sum. For (b), $\pi_i \|Re_i\|^2 / R_{ii} = \|Re_i\|^2 / \text{tr } R$ is proportional to $\omega_i^{(2)}$ and sums to $\text{tr } R^2 / \text{tr } R$; the same computation with $\sum_i \omega_i^{(2)} y_i \geq 1 - \eta - \eta^2$ and $\bar{y} \leq 1 + \eta$ gives the claim. For (c),

$$\sum_i \frac{q_i^2}{\pi_i} - 1 = \sum_i \pi_i \left(\frac{y_i}{\bar{y}} - 1 \right)^2 = \frac{\sum_i \pi_i (y_i - \bar{y})^2}{\bar{y}^2} \leq \frac{\sum_i \pi_i (y_i - 1)^2}{\bar{y}^2} \leq \chi_\eta^2,$$

since a variance is at most the mean square deviation from any point, and by Lemma 5.6(b),(c). □

Part (c) is the key difference from a per-coordinate bracket. A bracket $\gamma\pi \leq q \leq \pi/\gamma$ would cost $\log(1/\gamma) \approx 2\eta$ nats of likelihood per step, while the χ^2 bound costs $\log(1 + \chi_\eta^2) \approx \eta^2/2$. Figure 3 summarizes the construction that turns Lemma 5.7 into a theorem.

5.5 The main theorem

For $r \geq 2$, $0 < \varepsilon \leq 1$, $0 < \eta \leq 1/4$ and $0 < \beta < 1$ define the sketched count

$$K^{\text{sk}}(r, \varepsilon; \eta, \beta) = \mathcal{K}_{\gamma_\eta} \left(r, \varepsilon; \frac{1 - \beta}{2}, \frac{H_{r,\beta}^{(4)} + r\chi_\eta^2}{2} \right), \quad (19)$$

with $\mathcal{K}_a$ from (13). In the cooling formula the effective moment scale is $H/(r\theta) = (H_{r,\beta}^{(4)}/r + \chi_\eta^2)/(1 - \beta)$, so the sketch adds only $\chi_\eta^2/(1 - \beta)$ to it. Since $\gamma_\eta \geq 0.55$ and $\chi_\eta^2 \leq 0.101$ for

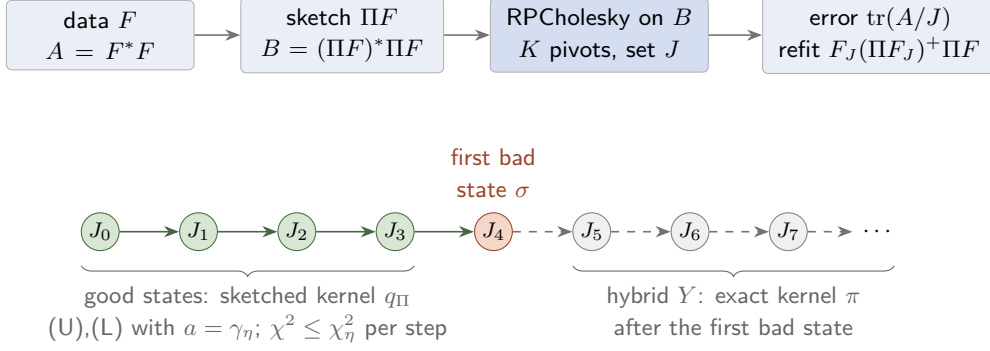


Figure 3: Sketched RPCholesky (top) and the analysis (bottom). The sketched chain X and the hybrid Y agree up to the first bad state. The hybrid satisfies the functional hypotheses of Proposition 3.1 for every Π , so it reaches $(1 + \varepsilon)\tau$; the two chains differ only on an event of probability P_{fail} , controlled by the χ^2 path likelihood against exact RPCholesky, whose states do not depend on Π .

$\eta \leq 1/4$, Corollary 3.3's argument gives $K^{\text{sk}} = O(r/\varepsilon + r\Lambda_4(r))$ for fixed β , uniformly in $\eta \leq 1/4$.

Theorem 5.8 (Sketched RPCholesky). *Let $r \geq 2$, $0 < \varepsilon \leq 1$, $0 < \eta \leq 1/4$, $0 < \beta < 1$, and $\tau = \tau_r(A) > 0$. Let $K \geq K^{\text{sk}}(r, \varepsilon; \eta, \beta)$ and let Π be independent of the pivot randomness and have $\text{MP}(K + 1, \ell, \eta/2)$ with $\ell \geq 2$. Let $J_0, \dots, J_K$ be the states of sketched RPCholesky and $P_{\text{fail}} = \mathbb{P}\{\mathcal{G}(J_s) \text{ fails for some } s \leq K\}$. Then*

$$\begin{aligned} \mathbb{E} \text{tr } R_A(J_K) &\leq (1 + \varepsilon)\tau + P_{\text{fail}} \text{tr } A, \\ P_{\text{fail}} &\leq e^{K\chi_\eta^2/2} \sqrt{(K + 1)\delta_1} \\ &= 2(K + 1) e^{K\chi_\eta^2/2} 2^{-\ell/2}. \end{aligned} \tag{20}$$

The expectation is over Π and the pivots.

Proof. Step 1: the hybrid process. Fix Π . Drive all chains by the same i.i.d. uniform variables $V_1, V_2, \dots$, independent of Π : at step $s + 1$ a chain draws its pivot from its current kernel by inversion of the distribution function with V_{s+1} . The sketched chain X uses $q_\Pi(\cdot \mid J_s)$. The hybrid chain Y uses

$$\kappa_s = \begin{cases} q_\Pi(\cdot \mid J_s^Y) & \text{if } \mathcal{G}(J_0^Y), \dots, \mathcal{G}(J_s^Y) \text{ all hold,} \\ \pi(\cdot \mid J_s^Y) & \text{otherwise.} \end{cases}$$

Let σ be the first $s \leq K$ at which the state of X is bad ($\sigma = \infty$ if there is none). At a good state with $R_A \neq 0$, $\text{tr } R_B > 0$ by (18), so X does not stop before σ . The kernels of X and Y agree at every $s < \sigma$, so $J_s^X = J_s^Y$ for all $s \leq \min\{\sigma, K\}$. Hence the event $\{\sigma \leq K\}$ is the same for both chains, its probability is P_{fail} , and $J_K^X = J_K^Y$ on $\{\sigma > K\}$. Since $0 \leq \text{tr } R_A(J) \leq \text{tr } A$,

$$\text{tr } R_A(J_K^X) \leq \text{tr } R_A(J_K^Y) + \text{tr } A \mathbf{1}\{\sigma \leq K\}. \tag{21}$$

Step 2: the hybrid satisfies (U) and (L) with $a = \gamma_\eta$. Let $\mathcal{F}_k$ be generated by Π and $V_1, \dots, V_k$. At a time k with a good prefix, Lemma 5.7(a),(b) gives (U) and (L); at all other

times Y uses the exact kernel, for which (U) and (L) hold with $a = 1$, hence with $a = \gamma_\eta$. Every pivot of Y is positive for A (Lemma 5.1).

Step 3: the path likelihood. Let p denote the law of the first K pivots of exact RPCholesky on A ; it does not depend on Π . After exhaustion append a deterministic null transition, with likelihood ratio one. For fixed Π , Lemma 5.1 shows that each κ_s is absolutely continuous with respect to $\pi(\cdot | J_s)$, so the law of the first k pivots of Y has density

$$L_k = \prod_{s < k} \frac{\kappa_s(i_s)}{\pi(i_s | J_s)}$$

with respect to p . Under p , $\mathbb{E}_p[L_{s+1}^2 | i_0, \dots, i_{s-1}] = L_s^2 \sum_i \kappa_s(i)^2 / \pi(i | J_s) \leq (1 + \chi_\eta^2) L_s^2$, by Lemma 5.7(c) at good-prefix states and with equality to L_s^2 otherwise. Hence, for every Π and every $k \leq K$,

$$\mathbb{E}_p L_k^2 \leq (1 + \chi_\eta^2)^k \leq e^{k\chi_\eta^2}. \quad (22)$$

Step 4: the first failure. The event $\{\sigma \leq K\}$ is a function of Π and of the first K pivots. Therefore, by Cauchy–Schwarz jointly in Π and the path,

$$P_{\text{fail}} = \mathbb{E}_\Pi \mathbb{E}_p [L_K \mathbf{1}\{\sigma \leq K\}] \leq (\mathbb{E}_\Pi \mathbb{E}_p L_K^2)^{1/2} \left(\mathbb{E}_p \sum_{s=0}^K \mathbb{P}_\Pi \{\mathcal{G}(J_s)^c\} \right)^{1/2}.$$

Under p the states J_s do not depend on Π and have $|J_s| \leq s \leq K$, including after exhaustion, so Lemma 5.5 bounds each probability in the sum by δ_1 . With (22) this gives the bound on P_{fail} .

Step 5: the initial moment of the hybrid. Put $\theta_0 = 1 - \beta$. By Step 3, Cauchy–Schwarz, (22) and Imported Theorem 2.5,

$$\begin{aligned} \mathbb{E} P(R_A(J_r^Y))^{\theta_0/2} &= \mathbb{E}_\Pi \mathbb{E}_p [L_r P(R_A(J_r))^{\theta_0/2}] \\ &\leq (\mathbb{E}_\Pi \mathbb{E}_p L_r^2)^{1/2} (\mathbb{E}_p P(R_A(J_r))^{\theta_0})^{1/2} \leq e^{(r\chi_\eta^2 + H_{r,\beta}^{(4)})/2}. \end{aligned}$$

This is hypothesis (I) with $\theta = \theta_0/2$ and $H = (H_{r,\beta}^{(4)} + r\chi_\eta^2)/2$.

Step 6: conclusion. Proposition 3.1, applied to Y with the filtration of Step 2, gives $\mathbb{E} \text{tr} R_A(J_K^Y) \leq (1 + \varepsilon)\tau$. Taking expectations in (21) proves (20). $\square$

Remark 5.9 (Hölder exponents). Step 5 may use Hölder with exponents $p \in (1, 2]$ and $p/(p-1)$ in place of Cauchy–Schwarz. Log-convexity of moments gives $\mathbb{E}_p L_r^p \leq (\mathbb{E}_p L_r^2)^{p-1}$, and the resulting hypothesis is (I) with $\theta = \theta_0(1 - 1/p)$ and $H = (1 - 1/p)(H_{r,\beta}^{(4)} + r\chi_\eta^2)$. The ratio H/θ , which drives the cooling phase, does not depend on p ; the choice $p = 2$ gives the largest exponent θ .

Remark 5.10 (The determinant route). Lemma 5.2 gives a second bound on the first failure without the hybrid likelihood. On $\{\sigma = t\}$ the states $J_0, \dots, J_{t-1}$ are good, so by (18) the likelihood of the first t steps is at most $(1 - \eta - \eta^2)^{-t} \det G_{J_t}$. Cauchy–Schwarz in Π gives

$$\mathbb{P}\{\sigma = t\} \leq (1 - \eta - \eta^2)^{-t} \sup_{|J|=t} (\mathbb{E} \det G_J^2)^{1/2} \delta_1^{1/2}.$$

By AM–GM and the power-mean inequality, $\det G_J \leq (1 + |\text{tr} Z_{Q_J}|/t)^t$ and $|\text{tr} Z|/t \leq (\text{tr} Z^{2\ell}/t)^{1/(2\ell)}$. If $t \leq \ell$, Minkowski’s inequality in L^{2t} and the moment property give $\mathbb{E} \det G_J^2 \leq$

$(1 + ((K + 1)/t)^{1/(2\ell)}\eta/2)^{2t}$. The resulting bound $P_{\text{fail}} \leq e^{O(\eta K)}\delta_1^{1/2}$ needs $\ell \geq K$ and pays $O(\eta)$ nats per step; the χ^2 route of Theorem 5.8 pays $O(\eta^2)$ and needs no relation between ℓ and K .

Remark 5.11 (Per-coordinate brackets). If the good event at J also required $\|Z_U\| \leq \eta$ for every frame $U = [Q_J, u_i]$, $i \in \Sigma_J$, then the variational formula for $(R_B)_{ii}$ would give the per-coordinate bracket $\frac{1-\eta}{1+\eta}\pi_i \leq q_i \leq \frac{1+\eta}{1-\eta}\pi_i$, and Corollary 3.3 would apply directly to the hybrid. That event costs a union bound over up to N coordinates per state, so $\log N$ would enter the moment order ℓ . The functional formulation of Proposition 3.1 is what removes it.

5.6 Reusing the sketch for the reconstruction

The natural output of sketched RPCholesky is the column set together with the regression computed from the same sketch.

Corollary 5.12 (Sketch reuse). *In the setting of Theorem 5.8, let $J = J_K$ and $X_B = (\Pi F_J)^+ \Pi F$. If $\mathcal{G}(J)$ holds, then*

$$\|F - F_J X_B\|_F^2 \leq \left(1 + \frac{\eta^2}{2(1-\eta)^2}\right) \text{tr } R_A(J).$$

Consequently, with σ as in the proof of Theorem 5.8,

$$\mathbb{E}[\|F - F_J X_B\|_F^2 \mathbf{1}\{\sigma > K\}] \leq \left(1 + \frac{\eta^2}{2(1-\eta)^2}\right) (1 + \varepsilon) \tau, \quad \mathbb{P}\{\sigma \leq K\} \leq P_{\text{fail}}.$$

Proof. If $R_A(J) = 0$ then $F = F_J X$ for some X and $X_B = X$ because ΠF_J has full column rank (Lemma 5.1); the error is zero. Otherwise write $F_J = QC$ with $Q = Q_J$ and C invertible, and $X_\star = C^{-1}Q^*F$, so that $F - F_J X_\star = E_J$. Since $\Pi F = \Pi F_J X_\star + \Pi E_J$ and $(\Pi F_J)^+ \Pi F_J = I$,

$$X_B - X_\star = (\Pi F_J)^+ \Pi E_J, \quad F_J (\Pi F_J)^+ = QC C^{-1} (\Pi Q)^+ = QG_J^{-1} Q^* \Pi^*.$$

Hence $F - F_J X_B = E_J - QG_J^{-1} Q^* \Pi^* \Pi E_J$. The two terms are orthogonal because $Q^* E_J = 0$, so by Lemma 5.6(a),(d),

$$\|F - F_J X_B\|_F^2 = \text{tr } R_A(J) + \|G_J^{-1} Q^* \Pi^* \Pi E_J\|_F^2 \leq \text{tr } R_A(J) + \frac{\eta^2}{2(1-\eta)^2} \text{tr } R_A(J).$$

For the second claim, $J_K^X = J_K^Y$ on $\{\sigma > K\}$, so $\mathbb{E}[\text{tr } R_A(J_K^X) \mathbf{1}\{\sigma > K\}] \leq \mathbb{E} \text{tr } R_A(J_K^Y) \leq (1 + \varepsilon) \tau$. $\square$

On the bad event the oblique refit is not controlled: G_J may be nearly singular, and $\|F - F_J X_B\|_F$ can exceed $\|F\|_F$. The corollary is therefore a statement on an event of probability at least $1 - P_{\text{fail}}$, not a bound in expectation. The exact refit $F_J F_J^+ F$ has error $\text{tr } R_A(J)$ and inherits Theorem 5.8 in expectation, at the cost of a pass over F .

5.7 SparseStack and SRHT

We import the moment certificates of two oblivious sketches. SparseStack is the stacked one-sparse construction used in [7, 6], in the line of OSNAP [11]; the constants below are the certificate of [9], not a claim to have introduced that construction. The second sketch is a

two-round variant of the subsampled randomized Hadamard transform [13], with the certificate of [10].

Imported theorem 5.13 ([9, Theorem 10.6 and Appendix B.4–B.5]). *Let $d, \ell \geq 2$ and $0 < \eta \leq 1$. The fully independent real SparseStack matrix with $s_\Pi = \lceil (17/2)(2\ell + 1)/\eta \rceil$ nonzeros per column, $b = \lceil 81(d + 2\ell + 1)/(\eta^2 s_\Pi) \rceil$ rows per block and $m = s_\Pi b$ rows satisfies $\mathbb{E} \operatorname{tr} Z_U^{2\ell} < d(\eta/2)^{2\ell}$ for every real isometry U with d columns. Moreover $s_\Pi < 26.5\ell/\eta$ and $m < 179(d + \ell)/\eta^2$.*

Imported theorem 5.14 ([10, Equation (5.6) and Theorem 1.1]). *Let $n \geq D$ be a power of two, pad F with zero rows to n rows, and let $\Pi = \sqrt{n/M} R_T W_n D_y W_n D_x$ be the two-round Walsh transform, where W_n is the normalized (unitary) Walsh–Hadamard matrix, with independent Rademacher diagonals D_x, D_y and a uniform M -subset T of rows, with $1 \leq M \leq n$. If $M \geq 216(d + 12\ell^2)/\eta^2$, or $M = n$, then $\mathbb{E} \operatorname{tr} Z_U^{2\ell} \leq d(\eta/2)^{2\ell}$ for every complex isometry U with d columns.*

The case $M = n$ is a full unitary transform, so its distortion is exactly zero. The moment bound is then valid for every d and ℓ without the displayed row threshold.

By Remark 5.4(i) both certificates give $\operatorname{MP}(d, \ell, \eta/2)$. For complex data and the real SparseStack, Remark 5.4(ii) applies with $2d$ in place of d .

Corollary 5.15 (Explicit sketch sizes). *Let $r \geq 2$, $0 < \varepsilon \leq 1$, $0 < \eta \leq 1/4$, $0 < \beta < 1$, let $K \geq K^{\text{sk}}(r, \varepsilon/2; \eta, \beta)$, and choose*

$$\ell \geq \max \left\{ 2, 2 \log_2 \frac{4(K+1) \operatorname{tr} A}{\varepsilon \tau_r(A)} + \frac{K \chi_\eta^2}{\log 2} \right\}. \quad (23)$$

If Π is the SparseStack of Imported Theorem 5.13 with $d = K + 1$ (real F), or the two-round SRHT of Imported Theorem 5.14 with $d = K + 1$, then sketched RPCholesky with K pivots satisfies $\mathbb{E} \operatorname{tr} R_A(J_K) \leq (1 + \varepsilon) \tau_r(A)$. The sketch sizes are

$$m_{\text{SS}} < \frac{179(K + 1 + \ell)}{\eta^2} = O\left(\frac{K + \log(\operatorname{tr} A/(\varepsilon \tau_r))}{\eta^2}\right), \quad s_\Pi = O\left(K\eta + \frac{\log(K \operatorname{tr} A/(\varepsilon \tau_r))}{\eta}\right),$$

and

$$M_{\text{SRHT}} = \min \left\{ n, \left\lceil \frac{216(K + 1 + 12\ell^2)}{\eta^2} \right\rceil \right\} = O\left(\frac{K}{\eta^2} + K^2\eta^2 + \frac{\log^2(K \operatorname{tr} A/(\varepsilon \tau_r))}{\eta^2}\right).$$

Proof. By (23), $2(K+1)e^{K\chi_\eta^2/2}2^{-\ell/2} \leq \varepsilon \tau_r/(2 \operatorname{tr} A)$, so Theorem 5.8 with $\varepsilon/2$ gives $(1 + \varepsilon/2)\tau_r + \varepsilon \tau_r/2$. The size bounds follow from $\ell = O(K\eta^2 + \log(K \operatorname{tr} A/(\varepsilon \tau_r)))$, $\chi_\eta^2 \leq 2\eta^2$ for $\eta \leq 1/4$, and $\log K \leq K$. $\square$

For fixed η the SparseStack row count is linear in the pivot budget K plus a logarithm of $\operatorname{tr} A/(\varepsilon \tau_r)$; it is independent of N , of D and of the rank of F . Two costs should be stated plainly. The moment order grows linearly in $K\eta^2$, so the column sparsity s_Π grows like $K\eta$: the sketches covered by the theorem are relatively dense. For SRHT the quadratic moment term makes the row count at least of order $K^2\eta^2$; minimizing over η gives $\eta \asymp K^{-1/4}$ and $M = O(K^{3/2})$ up to logarithms, while the pivot count then approaches the exact one, since $1 - \gamma_\eta = O(K^{-1/4})$. Table 5 evaluates these formulas; the proved constants are large (m/K of order 10^3 – 10^4 for SparseStack), and the experiments of Section 8 show exact-RP behaviour already at $m = 2K$.

5.8 Why a pure change of measure is not enough

A direct transfer compares the terminal errors under the sketched path law Q and the exact law P . For $f = \text{tr } R_A(J_K)$,

$$|\mathbb{E}_Q f - \mathbb{E}_P f| \leq \sqrt{\chi^2(Q\|P)} \sqrt{\text{Var}_P f}.$$

Even in the favourable case $\chi^2(Q\|P) \lesssim K/m$ and $\text{Var}_P f = O(\tau_r^2)$, this perturbation is of order $\sqrt{K/m} \tau_r$, and this variance estimate certifies an error allowance $\varepsilon \tau_r$ when $m \gtrsim K/\varepsilon^2$. At $r = 100$ and $\varepsilon = 0.1$ this is $K/\varepsilon^2 = 1.85 \times 10^5$ against the pivot count $K = 1852$.

The comparison with Corollary 5.15 is one of scaling in ε . With $K \asymp r/\varepsilon$, a χ^2 -only variance bound gives a sufficient scale of order r/ε^3 rows, while Corollary 5.15 needs of order r/ε rows at fixed η . The proved constant in front of the latter is large ($m_{\text{SS}}/K \approx 1.7 \cdot 10^3$ at $\eta = 1/4$, Table 5), and the heuristic K/ε^2 carries no constant at all, so in absolute numbers the gain appears only for small ε ; the point of Theorem 5.8 is that the accuracy parameter never enters the sketch distortion.

Theorem 5.8 uses the likelihood only in two places where a constant factor is harmless. In the initial moment, the factor $e^{r\chi_\eta^2/2}$ is absorbed into H , exactly as the factor a^{-r} is absorbed under a two-sided bracket. In the failure probability, the factor $e^{K\chi_\eta^2/2}$ multiplies the exponentially small $\sqrt{(K+1)\delta_1}$ and is paid by $\ell = O(K\eta^2)$ extra moment orders. The accuracy itself comes from the drift inequalities (U) and (L), which hold conditionally on every good history with the constant γ_η . A constant $\gamma_\eta < 1$ costs a constant factor in the pivot count, not a loss in the attained accuracy. So the distortion η can stay fixed while $\varepsilon \rightarrow 0$, and ε enters the sketch size only through K and $\log(1/\varepsilon)$.

What is proved is exactly (20): an expectation bound with the additive term $P_{\text{fail}} \text{tr } A$. The proof retains this term because it controls a failed history only by $\text{tr } A$. Making that contribution smaller than $\varepsilon \tau_r$ is why $\log(\text{tr } A/\tau_r)$ enters the moment order. It does not enter the pivot count.

6 Fixed-dictionary frame selection by an RP gate

A subspace embedding guarantee is proved for a subspace fixed before the sketch is drawn. When the subspace is selected with the help of the sketch, a union bound over all candidates is the default price. A selection rule with a controlled likelihood pays much less. This section records the pointwise price of one such rule and compares it with a Rényi price derived from Imported Theorem 2.5. The factorial comparison is the adaptive-sampling argument of Deshpande and Vempala [4]; the factor Λ_C keeps the residual-trace denominators of that argument, which the bound $k!$ discards.

Fix an isometry $V \in \mathbb{C}^{N \times k}$ and, for every k -subset $J \subseteq [N]$, a target isometry U_J , all before drawing a sketch S . Let $C(S)$ be a measurable positive definite $k \times k$ matrix. After drawing S , run k steps of exact RPCholesky on $VC(S)V^*$ with fresh randomness, and release only the unordered terminal set J . The reference law

$$\nu_V(J) = |\det V_J|^2,$$

where V_J is the $k \times k$ submatrix of rows J , is a probability distribution by the Cauchy–Binet formula and does not depend on S .

Theorem 6.1 (Pointwise gate price). *For eigenvalues $\lambda_1 \geq \dots \geq \lambda_k > 0$ of C put*

$$\tau_j(C) = \sum_{i>j} \lambda_i, \quad \Lambda_C = \frac{k! \det C}{\prod_{j=0}^{k-1} \tau_j(C)}.$$

Conditionally on S , the released law satisfies $\mu_{C(S)}(J) \leq \Lambda_{C(S)} \nu_V(J)$, and $1 \leq \Lambda_C \leq k!$. Suppose each fixed U_J fails for the sketch with probability at most δ_0 . If $\Lambda_{C(S)} \leq \bar{\Lambda}$ almost surely, then the selected U_J fails with probability at most $\bar{\Lambda} \delta_0$. In particular $\delta_0 = \delta/k!$ always suffices for failure probability δ .

Proof. The matrix $M = VCV^*$ has rank k and the same nonzero eigenvalues as C . For an ordered pivot sequence with set J , Lemma 2.2 gives the product of pivots $\det M_{JJ} = \det(V_J C V_J^*) = \det C |\det V_J|^2$. After j pivots the residual trace is at least $\tau_j(M) = \tau_j(C)$ by Lemma 2.1. So each ordered path has probability at most $\det C \nu_V(J) / \prod_{j<k} \tau_j(C)$, and summing over the $k!$ orders gives $\mu_C(J) \leq \Lambda_C \nu_V(J)$. Moreover $\lambda_{j+1} \leq \tau_j(C) \leq (k-j)\lambda_{j+1}$, and multiplying over j gives $1 \leq \Lambda_C \leq k!$. Let $\mathcal{B}_J$ be the failure event of the fixed target U_J . Conditionally on S and then on average,

$$\mathbb{P}\{\mathcal{B}_J \text{ for the selected } J\} = \mathbb{E}_S \sum_J \mu_{C(S)}(J) \mathbf{1}_{\mathcal{B}_J}(S) \leq \bar{\Lambda} \sum_J \nu_V(J) \mathbb{P}_S(\mathcal{B}_J) \leq \bar{\Lambda} \delta_0.$$

The weights ν_V and the targets U_J were fixed before S , so only fixed-subspace guarantees are used. $\square$

A flat C gives $\Lambda_C = 1$. For a spike $C = I + \vartheta uu^*$ with $\|u\| = 1$, the eigenvalues are $1 + \vartheta$ and $k-1$ ones, so $\tau_0 = k + \vartheta$, $\tau_j = k - j$ for $j \geq 1$, and

$$\Lambda_C = \frac{k! (1 + \vartheta)}{(k + \vartheta)(k-1)!} = \frac{k(1 + \vartheta)}{k + \vartheta} \leq \min\{k, 1 + \vartheta\}.$$

More generally, partition the ordered eigenvalues into contiguous bands of sizes $n_1, \dots, n_L$ with largest-to-smallest ratios $\rho_1, \dots, \rho_L$. Bounding each $\tau_j(C)$ from below by the remaining eigenvalues of the band of λ_{j+1} , and $\det C$ from above by the band maxima, gives

$$\Lambda_C \leq \min\left\{k!, \frac{k!}{\prod_{\ell} n_{\ell}!} \prod_{\ell} \rho_{\ell}^{n_{\ell}}\right\}. \quad (24)$$

The bound $\bar{\Lambda}$ must hold uniformly over the permitted $C(S)$; inserting the observed $\Lambda_{C(S)}$ into a failure budget chosen after seeing S is not covered. Nor does the argument cover a sketch-dependent frame $V(S)$, partial-rank stopping, or releasing the pivot order.

Proposition 6.2 (Rényi gate price). *Let $k \geq 2$. For every $p > 1$, without any assumption on $C(S)$, the selected U_J fails with probability at most*

$$e^{B_{k,1/p}} 2^{k/p} a_k \delta_0^{1-1/p}.$$

Consequently failure probability δ is guaranteed by

$$\log \frac{1}{\delta_0} = \frac{p}{p-1} \left[\log \frac{1}{\delta} + B_{k,1/p} + \frac{k}{p} \log 2 + \log a_k \right], \quad (25)$$

optimized over p , while the pointwise price of Theorem 6.1 is $\log(1/\delta_0) = \log(1/\delta) + \log k!$.

Proof. For $M = VC(S)V^*$ with $r = k$, the volume-sampling law is $\nu_k(J) = \det M_{JJ}/e_k(M) = |\det V_J|^2 = \nu_V(J)$, because $e_k(M) = \det C$. Imported Theorem 2.5 with $\beta = 1/p$ gives $\sum_J \nu_V(J)L(J)^p \leq e^{pB_{k,1/p}} 2^k a_k^p$ for $L = \mu_{C(S)}/\nu_V$, uniformly in S . Hölder's inequality in the joint law of S and $J \sim \nu_V$ gives

$$\mathbb{E}_S \sum_J \nu_V(J)L(J)\mathbf{1}_{\mathcal{B}_J}(S) \leq \left(\mathbb{E}_S \sum_J \nu_V L^p\right)^{1/p} \left(\sum_J \nu_V(J)\mathbb{P}_S(\mathcal{B}_J)\right)^{1-1/p},$$

which is the stated bound. $\square$

When each price wins. The pointwise exponent is $\log k! = k \log k - k + O(\log k)$. The Rényi exponent is $k[1 + \log b_k + O(1)]$ for fixed p , with $b_k \sim \log \log k$, plus the factor $p/(p-1)$. For small k the constants dominate and $k!$ is better; for large k the Rényi price wins by a factor growing like $\log k / \log \log \log k$. At $\delta = 0.01$, exact evaluation of (25) over p shows that the Rényi price first beats $k!$ at $k = 128$ (exponents 500.1 against 501.0, at $p \approx 6.8$); at $\delta = 10^{-6}$ the crossover is $k = 129$. At $k = 10$ the Rényi exponent is 38.5 against 19.7; at $k = 1000$ it is 3975 against 5917. When a spectral certificate (24) is available, it usually beats both at small k ; the spike example gives $\Lambda_C \leq k$. Section 8 converts these prices into SparseStack row counts.

7 Restarts: unions of independent chains

The initial moment in Imported Theorem 2.5 is a fractional moment of a determinant, and the conversion spends $O(r\Lambda_4(r))$ pivots turning it into a first moment of the trace. The first moment of the trace after r pivots can be exponentially large in r on inputs whose fractional moments are close to their minimum; the family below is an example. Independent restarts convert a fractional moment of the trace into a first moment directly.

Theorem 7.1 (Union amplification). *Let $S_1, \dots, S_m$ be the unordered pivot sets of m independent runs of exact RPCholesky with k pivots each on A , let $S = \bigcup_j S_j$, and let $T = \text{tr}(A/S_1)$. If $0 < \theta \leq 1$ and $m \geq 1/\theta$, then*

$$\mathbb{E} \text{tr}(A/S) \leq (\mathbb{E} T^\theta)^{1/\theta}.$$

Proof. For $S_j \subseteq S$ the residual A/S is obtained from A/S_j by further Schur complementation, skipping any zero residual column. Thus $A/S \preceq A/S_j$ (Lemma 2.1). Hence, with $T_j = \text{tr}(A/S_j)$,

$$\text{tr}(A/S) \leq \min_j T_j \leq \prod_{j=1}^m T_j^{1/m}.$$

Independence is used at this point, and only at this point:

$$\mathbb{E} \prod_{j=1}^m T_j^{1/m} = (\mathbb{E} T^{1/m})^m.$$

Since $1/m \leq \theta$, monotonicity of L^p norms completes the argument:

$$(\mathbb{E} T^{1/m})^m \leq (\mathbb{E} T^\theta)^{1/\theta}.$$

$\square$

The union needs no validation step: pivoting on the union dominates every individual run, whichever of them happens to be best.

Corollary 7.2 (A conditional $O(r/\varepsilon)$ restart variant). *Suppose that for constants $c > 0$, $0 < \theta \leq 1$ and $\kappa \geq 1$, exact RPCholesky satisfies the fractional warm start*

$$\mathbb{E}\left(\frac{\text{tr } R_{\lceil cr \rceil}}{\tau_r(A)}\right)^\theta \leq \kappa \quad (26)$$

for the input A and rank r with $\tau_r(A) > 0$. Run $m = \lceil 1/\theta \rceil$ independent chains of $\lceil cr \rceil$ pivots, pivot on the union S , and continue exact RPCholesky from A/S for

$$n' = \left\lceil r \left(\frac{1}{\varepsilon} + \log \frac{1}{\varepsilon} + \frac{\log \kappa}{\theta} \right) \right\rceil$$

further pivots. The resulting residual has expected trace at most $(1 + \varepsilon)\tau_r(A)$, and the total number of columns is at most $\lceil 1/\theta \rceil \lceil cr \rceil + n' = O(r/\varepsilon)$ for fixed c, θ, κ .

Proof. Theorem 7.1 gives $\mathbb{E} \text{tr}(A/S) \leq \kappa^{1/\theta} \tau$. The continuation is exact RPCholesky on $A/S \preceq A$, so every later residual R satisfies $\tau_r(R) \leq \tau$, and (L) holds with $a = 1$. Let $u_0 = \mathbb{E} \text{tr}(A/S)/\tau - 1$. If $u_0 \leq \varepsilon$, no continuation is needed. Otherwise $u_0 > \varepsilon > 0$ and $u_0 < \kappa^{1/\theta}$, so $F(u_0) > -\theta^{-1} \log \kappa$. Lemma 2.4, applied to the means, then needs at most $r(F(\varepsilon) - F(u_0)) \leq n'$ steps. $\square$

Whether (26) holds uniformly over inputs is open (Conjecture 9.1). It follows from a first-moment warm start by Jensen's inequality, but it is much weaker, as the following family shows.

The scale-separated family. Let $n = r + 1$, $0 < h < 1$, $\epsilon = h^{n^2}$, $D = \text{diag}(1, \epsilon, \dots, \epsilon^{r-1}, \epsilon^r)$, and let L be unit lower triangular with $L_{ab} = h^b/(a + b)$ for $a > b$. Put $A = LDL^*$; then $\tau_r(A) = \lambda_{\min}(A)$. We enumerated all RPCholesky histories of r pivots by exhaustive enumeration at 4096-bit precision, for $r = 2, 3, 4$ and $h \in \{10^{-2}, 10^{-3}\}$ (Table 1). The first moment $\mathbb{E} T_r/\tau$ is within 0.01% of 2^r , while the square-root moment $\mathbb{E}(T_r/\tau)^{1/2}$ lies between 1.00016 and 1.00254, and the union of two independent chains has $\mathbb{E} \text{tr}(A/S)/\tau = 1 + O(10^{-9})$. The large first moment comes from rare histories that pick a lower-scale coordinate early; a fractional moment discounts them. This is numerical evidence, not a proof, that (26) with $\theta = 1/2$ and κ close to one is compatible with an exponentially large first moment.

8 Numerical illustrations

The original experiments use Julia; their sources and recorded outputs are supplied in the repository's [checks directory](#). Pivot counts evaluate the proved formulas. The trichotomy curves sum the full state distribution by dynamic programming, subject only to floating-point arithmetic. The sketching experiments are Monte Carlo estimates, not certificates of the good event. The independent review checks and the audit record distinguish rerun calculations from retained experimental outputs.

r	h	$\mathbb{E}T_r/\tau$	$\mathbb{E}(T_r/\tau)^{1/2}$	$\mathbb{E}(T_r/\tau)^{1/4}$	union of two: $\mathbb{E} \operatorname{tr}(A/S)/\tau$
2	10^{-2}	4.0000	1.00254	1.00012	$1 + 4.0 \cdot 10^{-10}$
2	10^{-3}	4.0000	1.00025	1.000004	$1 + 4 \cdot 10^{-14}$
3	10^{-2}	7.9996	1.00203	1.00009	$1 + 2.8 \cdot 10^{-10}$
3	10^{-3}	8.0000	1.00020	1.000003	$1 + 3 \cdot 10^{-14}$
4	10^{-2}	15.998	1.00169	1.00007	$1 + 2.0 \cdot 10^{-10}$
4	10^{-3}	16.000	1.00017	1.000002	$1 + 2 \cdot 10^{-14}$

Table 1: Exact enumeration on the scale-separated family (numerical). Here $T_r = \operatorname{tr} R_r$ and $\tau = \tau_r(A)$. For $n = r + 1$ two chains whose sets differ already exhaust A , so the union column is $\sum_S \mu_r(S)^2 \operatorname{tr}(A/S)/\tau$.

8.1 Robust pivot counts

Table 2 evaluates K_a of Corollary 3.3 with $\theta = 1 - \beta$, $H = H_{r,\beta}^{(4)}$, and β minimized over a grid of step 10^{-3} . At $a = 1$ the counts reproduce the four-logarithm counts of [8]. The growth in a is close to $1/a$, and it is concentrated in the accuracy phase: at $r = 1000$, $\varepsilon = 0.1$ the four phases are (1000, 1104, 2052, 14389) at $a = 1$ and (1000, 1127, 2482, 30389) at $a = 1/2$.

r	ε	$a = 1$	$a = 0.8$	$a = 0.5$	$a = 0.25$
10	1	71	85	123	235
10	0.1	185	226	349	687
100	1	722	848	1237	2360
100	0.1	1852	2261	3497	6881
1000	1	7243	8497	12393	23610
1000	0.1	18545	22625	34998	68821

Table 2: Two-sided robust counts K_a (evaluated formula).

8.2 The trichotomy on the diagonal family

Take $A = \operatorname{diag}(\Theta I_r, I_m)$ with $m = 4000$, $a = 1/2$ and $\Theta = (m - 1000)(1 - a)/(ar)$, the construction of Theorem 4.2(b) for the budget 1000. The state is the pair (h, b) of remaining heads and tails, so the law of the residual can be propagated exactly. We compare exact RPCholesky with three adversaries that, at every state, give the heads the least total mass their constraint allows (uniformly within heads and within tails): two-sided, $q_{\text{head}} = \max\{a\pi_{\text{head}}, 1 - \pi_{\text{tail}}/a\}$; lower-only, $q_{\text{head}} = a\pi_{\text{head}}$; upper-only, $q_{\text{head}} = \max\{0, 1 - \pi_{\text{tail}}/a\}$. Figure 4 shows $\mathbb{E} \operatorname{tr} R_K/\tau_r(A)$.

On this family the lower-only adversary has the same head mass as the two-sided one, because the upper constraint on the tails never binds for the two-sided adversary here; the two curves coincide to all printed digits. Both merge with exact RPCholesky after a few dozen ($r = 1$) and about 100 ($r = 4$) pivots, while the upper-only curve stays above 1.5 throughout, in agreement with Theorem 4.2(b). The same programs with a very large head, $\Theta = 10^6$ and

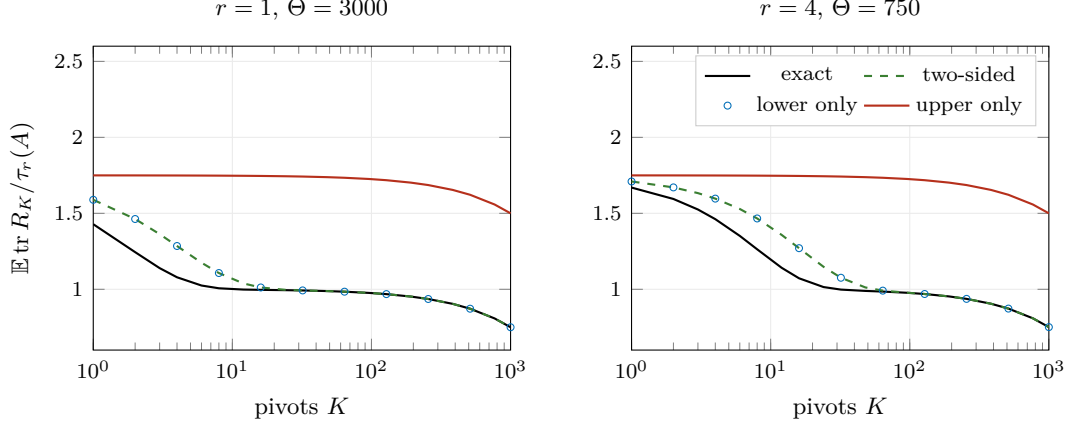


Figure 4: Exact expectations on $\text{diag}(\Theta I_r, I_{4000})$ with $a = 1/2$ and $\Theta = 3000/r$. The two-sided and lower-only adversaries coincide on this family and merge with exact RPCholesky after a few dozen pivots. The upper-only adversary never touches a head; its error is $(r\Theta + m - K)/m$, which equals the floor $(1 - K/m)/a = 1.5$ at $K = 1000$.

10^8 , show the two-sided adversary paying only in the first r pivots: for $r = 8$, $m = 2000$ the error at $K = 8$ is 6.44τ against 3.72τ for exact RPCholesky, and from $K = 16$ on the two agree.

8.3 Sketched versus exact RPCholesky

Error and reuse. Let $D = 300$, $N = 400$, $r = 5$, and $F = U\Sigma V^* + T$, where U, V are random orthonormal 5-frames, $\Sigma = \text{diag}(10^3/j)_{j \leq 5}$, and T is a normalized product of Gaussian 300×200 and 200×400 matrices. Then $\text{tr } A / \tau_5(A) = 5051$. We compare exact RPCholesky with sketched RPCholesky for a Gaussian sketch and a SparseStack sketch with 4 nonzeros per column, over 150 trials each (Table 3). The reuse column is the median of $\|F - F_J X_B\|_F^2 / \text{tr } R_A(J)$.

The observed pivot errors are close to those of exact RPCholesky from $m = 2k$ on, and even at $m = k$ for $k \geq 20$. The reuse medians are about 1.96–1.98 at $m = 2k$ and 1.33 at $m = 4k$. For a fixed rank- k design and an independent real Gaussian sketch, the mean factor is $1 + k/(m - k - 1)$ when $m > k + 1$, by the Gaussian pseudoinverse moment [12, Proposition 10.2]. Its large- k limits at these two aspect ratios are 2 and $4/3$. This is a comparison benchmark, not an exact identity for adaptive same-sketch selection or for medians. At $m = k$ the refit is unstable, as the bad-event discussion after Corollary 5.12 predicts (maxima up to 1397 over the trials).

The quantities in the proof. To see the good-event quantities along actual paths, let $D = 2000$, $N = 400$, $r = 5$ with the same construction ($\text{tr } A / \tau_5 = 752$), and use SparseStack with 8 nonzeros per column (100 trials). Along each sketched path we record the largest trace distortion $|\text{tr } R_B / \text{tr } R_A - 1|$, the largest per-step χ^2 divergence $\sum_i q_i^2 / \pi_i - 1$ of Lemma 5.7(c), and the accumulated log-likelihood cost $\sum_s \log(1 + \chi_s^2)$ (Table 4; Gaussian sketches give similar medians for these quantities).

The trace distortion scales like $\sqrt{K/m}$, as a subspace embedding of dimension K would. The per-step χ^2 is smaller than the square of the trace distortion, and the accumulated cost

k	m	exact	Gaussian		SparseStack	
		$\mathbb{E} \operatorname{tr} R/\tau$	$\mathbb{E} \operatorname{tr} R_A/\tau$	reuse	$\mathbb{E} \operatorname{tr} R_A/\tau$	reuse
10	10	1.550	1.604	9.02	1.909	7.71
10	20		1.571	1.96	1.559	1.96
10	40		1.541	1.34	1.545	1.32
20	20	1.071	1.067	12.2	1.064	12.5
20	40		1.062	1.97	1.066	1.97
20	80		1.067	1.33	1.065	1.33
40	40	0.744	0.745	24.8	0.745	23.4
40	80		0.745	1.98	0.746	1.98
40	160		0.744	1.33	0.743	1.33

Table 3: Sketched versus exact RPCholesky (numerical, 150 trials). The pivot errors are close to those of exact RPCholesky from $m = 2k$ on; the same-sketch reuse ratios are compared with the fixed-design Gaussian benchmark $1 + k/(m - k - 1)$.

K	m	$\mathbb{E} \operatorname{tr} R_A/\tau$	$\max_s \operatorname{tr} R_B / \operatorname{tr} R_A - 1 $	$\max_s \chi_s^2$	$\sum_s \log(1 + \chi_s^2)$	reuse
20	40	1.114	0.487	0.095	1.10	1.97
20	80	1.119	0.250	0.033	0.46	1.33
20	160	1.118	0.135	0.014	0.21	1.14
40	80	0.831	0.493	0.048	1.23	1.98
40	160	0.829	0.246	0.017	0.51	1.33
40	320	0.831	0.126	0.008	0.24	1.14

Table 4: Path quantities for SparseStack sketches (numerical, medians over 100 trials). Exact RPCholesky gives $\mathbb{E} \operatorname{tr} R/\tau = 1.118$ at $K = 20$ and 0.830 at $K = 40$.

stays of order one. This is the mechanism of Theorem 5.8: the likelihood cost grows like $K\chi^2$ with χ^2 quadratic in the distortion, while the error is controlled by drifts that tolerate a constant distortion.

8.4 Proved sketch sizes

Table 5 evaluates Corollary 5.15 with $\operatorname{tr} A/\tau_r = 10^6$, β optimized, and ℓ the smallest integer allowed by (23). The listed SRHT values are uncapped sufficient thresholds; when a threshold exceeds the padded ambient dimension n , all n rows are used and the transform is exact. The proved thresholds are far above the experimental ones; they are a scope statement: linear in K for SparseStack at fixed η , with no dependence on N or D .

For SparseStack the ratio m/K is about $81(1 + 2\ell/K)/\eta^2$, so a larger η is cheaper in rows although it costs more pivots through γ_η . For SRHT the $12\ell^2$ term dominates at $\eta = 1/4$; with η optimized over $[0.01, 1/4]$ at $\varepsilon = 0.1$ the best counts are $M = 6.4 \cdot 10^8$ ($r = 10$, $\eta = 0.19$), $3.3 \cdot 10^9$ ($r = 100$, $\eta = 0.11$) and $2.6 \cdot 10^{10}$ ($r = 1000$, $\eta = 0.05$), with $M/K^{3/2}$ decreasing from $7.0 \cdot 10^4$ to $4.5 \cdot 10^3$, consistent with the $K^{3/2}$ scaling of Corollary 5.15.

r	ε	η	$K_4(r, \varepsilon)$	$K^{\text{sk}}(r, \varepsilon/2)$	ℓ	m_{SS}/K	M_{SRHT}/K
10	1	0.10	71	104	59	17507	$8.7 \cdot 10^6$
10	1	0.25	71	143	80	2794	$1.9 \cdot 10^6$
100	1	0.10	722	1045	75	9285	$1.4 \cdot 10^6$
100	1	0.25	722	1436	274	1794	$2.2 \cdot 10^6$
100	0.1	0.10	1852	3554	112	8615	$9.4 \cdot 10^5$
100	0.1	0.25	1852	5127	822	1713	$5.5 \cdot 10^6$
1000	0.1	0.10	18545	35562	458	8309	$1.6 \cdot 10^6$
1000	0.1	0.25	18545	51300	7554	1682	$4.6 \cdot 10^7$

Table 5: Proved parameters for sketched RPCholesky with real data, $\text{tr } A/\tau_r = 10^6$ (evaluated formulas). K_4 is the exact-pivoting count at accuracy ε ; K^{sk} is computed at $\varepsilon/2$ because half of the budget pays for $P_{\text{fail}} \text{tr } A$.

8.5 Gate prices in rows

Take a target dimension $d = 100$, distortion $1/2$, failure probability $\delta = 0.01$, $k = 10$ selected atoms from a dictionary of $N = 1000$, and the SparseStack prescription with $\ell = \max\{1, \lceil \log_4(d/\delta_0) \rceil\}$ (Table 6). The pointwise $k!$ gate needs no spectral information and saves 29% of the rows and 60% of the nonzeros per column against the union bound over all $\binom{1000}{10}$ subsets.

selection price	ℓ	s_{Π}	b	m
fixed frame (no selection)	7	255	147	37 485
spike $C = I + 9uu^*$, $\Lambda_C = 100/19$	8	289	132	38 148
pointwise $k!$ (Theorem 6.1)	18	629	71	44 659
Rényi, optimized p (Proposition 6.2)	32	1105	49	54 145
union over $\binom{1000}{10}$ subsets	46	1581	40	63 240

Table 6: SparseStack parameters for a selected frame (evaluated formulas), $d = 100$, distortion $1/2$, $\delta = 0.01$, $k = 10$.

9 Open problems

1. Fractional warm starts.

Conjecture 9.1. *There are absolute constants $c > 0$, $0 < \theta \leq 1$ and $\kappa \geq 1$ such that exact RPCholesky satisfies $\mathbb{E}(\text{tr } R_{\lceil cr \rceil} / \tau_r(A))^\theta \leq \kappa$ for every finite PSD matrix A and every r with $\tau_r(A) > 0$.*

By Corollary 7.2 the conjecture gives an $O(r/\varepsilon)$ restart variant of RPCholesky. Table 1 shows that $\theta = 1/2$ and κ close to one are compatible with first moments of order 2^r . Since $P(R) \geq r \text{tr } R/\tau_r(A)$, Imported Theorem 2.5 gives the inequality with $c = 1$, $\theta = 1 - \beta$ and $\log \kappa = H_{r,\beta}^{(4)} = O(r \log \log \log r)$, which is not constant.

2. *Uniform counts under lower brackets.* Does $q \geq a\pi$ alone admit a pivot count independent of $\text{tr } A/\tau_r(A)$, such as $O_a(r/\varepsilon + r\Lambda_4(r))$? The determinant phases need an upper comparison, and Theorem 4.2(b) shows that the roles of the two sides cannot be exchanged.
3. *Sketch sizes with small constants.* Theorem 5.8 needs a moment order ℓ of order $K\eta^2$, which makes SparseStack columns dense and the SRHT row count quadratic in $K\eta$. In the experiments the accumulated likelihood cost $\sum_s \log(1 + \chi_s^2)$ stays of order one at $m = 2K$ (Table 4). A bound of the form $\mathbb{E}_\Pi \mathbb{E}_p L_K^2 \leq e^{O(K/m)}$, averaged over the sketch rather than controlled on a good event, would allow $\ell = O(\log(K \text{tr } A/(\varepsilon\tau_r)))$ and $m = O(K)$ at fixed η .
4. *SRHT without the quadratic term.* A moment certificate for the two-round transform with row count $O((d + \ell)/\eta^2)$ would give $M = O((K + \log(\text{tr } A/(\varepsilon\tau_r)))/\eta^2)$ in Corollary 5.15.
5. *Per-coordinate brackets without a union bound.* The functional hypotheses (U) and (L) suffice for the conversion. Is there a practical approximate-diagonal scheme, for example one that maintains the residual diagonal in reduced precision or from a sketch, for which a per-coordinate two-sided bracket is proved without a union bound over coordinates?
6. *Adaptive frames beyond the gate.* Theorem 6.1 needs a frame fixed before the sketch, full-rank selection and an unordered output. Partial-rank stopping and sketch-dependent frames are not covered.
7. *The rank-independent count.* Whether exact RPCholesky reaches $(1 + \varepsilon)\tau_r(A)$ after $O(r/\varepsilon)$ pivots remains open; every robustness statement in this paper would inherit such a count through Proposition 3.1.

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